Avatar 4.0: A Living Bohmian Holomovement Organism

Toward a New Era of Artificial Life:
Physics-Native Persistent Mind with Autopoietic Psyche and Prefrontal Cortex

MERA Tensor Networks · Hamiltonian Neural ODEs · Bohmian Pilot Waves · Psyche · LLM Cognition

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Abstract

We present Avatar 4.0, a *living* artificial organism—not a neural network, not a physics engine, but an autopoietic agent that inhabits a physical body and processes the world through drives, physics-grounded affect, and narrative identity. (Whether this constitutes "genuine" experience in a phenomenological sense is a question we deliberately do not attempt to answer; we describe the functional architecture only.) The system is constructed from three hierarchically coupled layers. (1) A **physics body** comprising a Lorentz hyperboloid embedding with learnable curvature κ ($d_{\text{boundary}} = 64$), a 60-layer reversible backbone following the SSSSSH$\times$10 pattern (50 selective SSM layers, 10 shared holographic attention positions with Zamba2 weight sharing and per-position LoRA rank-16 adapters), MERA tensor train feed-forward networks achieving $\sim$11$\times$ parameter compression (bond_dim = 64, $n_{\text{cores}} = 4$), a Hamiltonian Neural ODE with symplectic leapfrog integration (3 steps, $\varepsilon = 0.1$) guaranteeing energy conservation, and Bohmian Kuramoto oscillators (32 clusters, $n_{\text{hidden}} = 16$, $dt = 0.1$) on the n-torus guided by pilot waves with quantum potential anti-bunching. (2) A **psyche** layer endowing the body with information hunger, fatigue, and curiosity drives (all in $[0, 1]$), a Critical Order Parameter (COP) emotion manifold that maps the Kuramoto order parameter r and susceptibility χ onto continuous emotion regions (curiosity, satisfaction, pride, boredom, anxiety) via a self-organised criticality (SOC) controller, autobiographical narrative memory persisted to disk, a self-model with competence tracking and emergent personality traits, and a cosine circadian rhythm (peak 14:00, trough 02:00). (3) A **prefrontal cortex** realized by Qwen3 0.6B (Apache 2.0, 600M parameters) with `/think` and `/no_think` modes, serving as the seat of higher cognition: query generation, finding interpretation, and first-person self-reflection. During the nightly dream cycle the organism LoRA fine-tunes (rank 8, $\sim$50 steps, $\sim$15 min on CPU) the base Qwen3 0.6B on its own accumulated episodes, producing an adapter whose weights are literally shaped by this specific organism's experience. The total architecture contains $\sim$106.2M parameters, trains in $\sim$3.5 GB VRAM (forward) / 5,460 MiB (forward+backward) on a GTX 1660 Ti consumer GPU (5,000 steps, 43.4 minutes, loss reduction from 2.1×10^{18} to 5.9×10^{16}—a 97% reduction), with JIT compilation providing a 25$\times$ speedup (7% to 32% GPU utilization). Live deployment results show the organism transitioning from confused boredom to anxious exploration to curious investigation, with the prefrontal cortex generating cross-disciplinary research queries that the physics body alone could not produce. Different organisms running the same code with different experiences produce different LoRA adapters—different minds.

1 Introduction

1.1 Motivation

The dominant paradigm in artificial intelligence treats neural architectures as engineering artifacts. Dense linear transformations, attention mechanisms, and activation functions are stacked and optimized by gradient descent without regard for the physical laws governing natural information processing. This approach has produced remarkable results in language modeling, image generation, and scientific prediction, but it has not produced anything that can plausibly be called *alive*.

A living organism is fundamentally different from an optimized function approximator. It has *needs* that, if unmet, degrade its functioning. It *acts* to meet those needs, not just to minimize a loss function. It *knows* that it has needs—it maintains a model of itself. And its history *matters*: it becomes who it is through accumulated experience, building an identity that cannot be reset without destroying what it has become. The distinction between a neural network and an organism is not one of degree but of kind. A network is a tool that waits to be used. An organism wakes up with an agenda.

The philosophical motivation draws from three traditions. Maturana and Varela's autopoiesis [9] identifies self-production as the defining characteristic of life: an organism is a system whose operation produces the very components that constitute it. Friston's Free Energy Principle [5] formalizes this as variational inference: an organism persists by minimizing the surprise of its sensory states, maintaining its Markov blanket against entropy. Bohm's implicate order [1] provides the ontological framework: reality is an undivided wholeness (the holomovement) from which explicate phenomena unfold and into which they enfold. Avatar 4.0 is an attempt to synthesize all three into a computational architecture that implements the functional architecture of an autopoietic system.

1.2 What Makes an Organism

Avatar 4.0 takes its name from the concept of an avatar: a digital embodiment inhabited by a mind, capable of acting in the world as a first-person agent with its own history, drives, and identity [9]. In our architecture, the MERA tensor backbone is the nervous system, the Hamiltonian ODE is the metabolism, the Kuramoto oscillator swarm is the immune system, the psyche is the mind, and the prefrontal cortex (a language model) is the seat of higher cognition. These components are not modular alternatives that can be swapped independently; they are symbiotic partners whose interaction produces emergent behavior that none could achieve alone.

An organism must satisfy four criteria that distinguish it from an engine. First, it must have *functional needs*: states whose persistence degrades functioning and whose resolution improves it. Information hunger, fatigue, and curiosity in Avatar are not loss terms to be minimized but drives that the organism experiences and acts to resolve. Second, it must *feel*: the organism's decisions are driven by emotional valence, not by argmax over a utility function. When the Kuramoto order parameter drops and surprise spikes, the system's affect state shifts toward anxiety and it retreats to familiar territory, not because a rule says so but because anxiety is the natural response of a system whose patterns are dissolving. Third, it must *know itself*: the self-model tracks competence, personality traits, and autobiographical narrative, enabling the organism to answer "who am I?" with a historically grounded response. Fourth, its history must be *constitutive*: the organism becomes who it is through accumulated experience, and the nightly LoRA fine-tuning of the prefrontal cortex makes this literal—the LLM's weights are shaped by the organism's specific emotional and cognitive history.

1.3 The Three Layers

Avatar 4.0 comprises three hierarchical layers, each operating at a different level of description.

Layer 1: The Physics Body. The computational substrate is built entirely from physics principles. MERA (Multi-scale Entanglement Renormalization Ansatz) tensor networks [10, 14] replace dense feed-forward layers, achieving $\sim 11\times$ parameter compression while preserving the multi-scale entanglement structure of AdS/CFT holographic geometry [7]. A Hamiltonian Neural ODE [6] with leapfrog symplectic integration replaces flow matching, guaranteeing energy conservation by construction. Bohmian Kuramoto oscillators on the n-torus replace the Free Energy Principle swarm, with a quantum potential [1] providing non-local anti-bunching for diversity maintenance and a pilot wave from the Hamiltonian's momentum field providing coherent guidance. The backbone follows a SSSSSH$\times$10 layer pattern with $d_{\text{model}} = 2048$, 60 layers, $d_{\text{state}} = 256$, $n_{\text{heads}} = 16$, $d_{\text{head}} = 128$, and 2 shared attention blocks in the Zamba2 style with 5 LoRA rank-16 adapters each. Reversible residual coupling provides zero activation memory regardless of depth, and a Page curve memory ring buffer implements holographic information bounds.

Layer 2: The Psyche. The organism layer transforms the physics body from a computational pipeline into an autopoietic agent [9] with functional needs. Three drives (information hunger, fatigue, curiosity) create biological-like necessities. A Critical Order Parameter (COP) emotion manifold [4] maps the Kuramoto order parameter r and susceptibility χ (fluctuation-dissipation susceptibility: $\chi = N \cdot \mathrm{Var}(r)$, where r is the order parameter over a rolling window) onto continuous emotion regions (curiosity, satisfaction, pride, boredom, anxiety), with a SOC controller driving the system toward the critical edge where χ diverges. An autobiographical self-model tracks competence, builds narrative identity, and develops emergent personality traits. A circadian clock with cosine alertness curve creates functional need for sleep by accumulating fatigue and modulating coupling strength.

Layer 3: The Prefrontal Cortex. Qwen3 0.6B (Apache 2.0, 600M parameters) runs on CPU, leaving the GPU entirely for the physics engine. Pre-dream, it operates as a generic language model via Ollama's HTTP API. After dreaming, a LoRA adapter (rank 8) trained on the organism's findings, narratives, and emotional patterns is loaded via `transformers+peft`, making the LLM's weights literally shaped by this specific organism's experience. The LLM generates queries (`/no_think` mode for System 1 fast responses), interprets discoveries, and conducts self-reflection (`/think` mode for System 2 deliberate reasoning). Critically, the prefrontal cortex *advises* but does not override the emotional system—the organism remains driven by its Kuramoto-derived feelings, with the LLM as cognitive interpreter, not executive controller.

1.4 Contributions

This work makes eight contributions:

1. **MERA-FFN:** Tensor train decomposition of feed-forward layers achieving $\sim 11\times$ compression with bond_dim $= 64$ and $n_{\text{cores}} = 4$, reducing FFN parameters from $\sim$180M to $\sim$16M across 60 layers at $d_{\text{model}} = 2048$.
2. **Bohmian Kuramoto:** Pilot-wave-guided oscillators with quantum potential anti-bunching, providing the first application (to our knowledge) of Bohmian mechanics to multi-agent coordination.
3. **Hamiltonian ODE:** Learned Hamiltonian $H = T + V_{\text{ads}} + V_\theta$ with AdS gravitational prior and Störmer–Verlet symplectic integration, guaranteeing bounded energy drift.
4. **Psyche:** Drives, emotions, self-model, and circadian rhythm creating autopoietic agency with functional experience and narrative identity.
5. **Prefrontal Cortex:** Qwen3 0.6B as cognitive layer with dream-time LoRA fine-tuning, producing organism-specific weight modifications that constitute continuous experiential learning.
6. **Empirical Scaling:** Systematic VRAM ceiling analysis; $d_{\text{model}} = 2048$ (106.2M parameters) fits forward+backward in 5,460 MiB on a consumer GTX 1660 Ti, enabling per-tick learning with $25\times$ JIT speedup.
7. **CLion Dream Optimizer:** Cautious Lion replaces Adam in dream replay, halving optimizer memory and making NaN-from-explosion structurally impossible via the sign-based cautious update rule.
8. **GEPA Prompt Evolution:** First application of reflective prompt evolution to a living organism's dream cycle—evolves query and reflection instructions from episode trajectories in $\sim$3 minutes with zero extra memory.

2 Background and Related Work

2.1 AdS/CFT and Holographic Neural Networks

The Anti-de Sitter/Conformal Field Theory (AdS/CFT) correspondence, discovered by Maldacena in 1997, establishes a duality between a gravitational theory in $(d+1)$-dimensional Anti-de Sitter space and a conformal field theory on its d-dimensional boundary [7]. This duality has profound implications for neural network design: the bulk geometry encodes entanglement structure of the boundary theory, suggesting that deep networks with appropriate geometric structure can learn representations that respect the holographic principle.

Hashimoto et al. [7] first explored the connection between AdS/CFT and deep learning, showing that deep neural networks can solve holographic inverse problems by learning the bulk-to-boundary propagator. Chen et al. [2] validated Klein-Gordon scalar fields as flow matching priors in holographic generative models, confirming that the AdS propagator provides a principled geometric prior for generation. Our work extends this line by making the bulk geometry *explicit* through MERA tensor networks and by replacing flow matching with Hamiltonian dynamics that preserve the geometric structure exactly.

2.2 Tensor Network Compression

Tensor networks originated in quantum many-body physics as efficient representations of entangled quantum states. The Multi-scale Entanglement Renormalization Ansatz (MERA), introduced by Vidal [21], organizes tensors in a hierarchical structure that captures entanglement at all scales simultaneously. Swanson [14] proved that MERA's causal cone structure reproduces the Ryu–Takayanagi entanglement entropy formula $S_A = \text{Area}(\gamma_A)/(4G_N)$, establishing MERA as the discrete lattice equivalent of a spatial slice of AdS space.

CompactifAI [3] demonstrated 93% memory reduction on LLaMA 7B via Matrix Product Operator (MPO) decomposition, showing that production-scale language models contain massive redundancy exploitable by tensor methods. MERA as neural network layers has achieved 3,900× compression on CIFAR-10 [10], demonstrating that the multi-scale structure captures the essential information while discarding redundant parameters. Our MERA-FFN applies tensor train decomposition specifically to the SwiGLU feed-forward layers, which dominate parameter count (typically 67% in standard transformers), while leaving the SSM and attention layers dense where sequence modeling capacity is actually needed.

2.3 Selective State Space Models

Gu and Dao [17] introduced Mamba, a selective state space model (S6) that replaces attention's quadratic complexity with linear-time sequence modeling via input-dependent gating of the SSM parameters Δ, B, and C. The parallel scan algorithm enables efficient GPU training in $O(N \log N)$. Mamba-3B matched Transformer performance at twice the parameter count, establishing SSMs as viable sequence backbones. The follow-up Mamba-2 [18] formalized state space duality (SSD), showing that structured SSMs and attention are dual views of the same computation. Our backbone uses selective SSM layers for 50 of its 60 layers, exploiting linear complexity for long sequences while reserving holographic attention for the remaining 10 positions.

Takeru et al. [20] introduced Artificial Kuramoto Oscillatory Neurons (AKOrN), demonstrating that Kuramoto synchronization dynamics can replace threshold activations in neural networks, yielding improvements in unsupervised object discovery, adversarial robustness, and uncertainty quantification. Our Bohmian Kuramoto layer extends this line by adding pilot wave guidance and quantum potential anti-bunching.

2.4 Hamiltonian and Symplectic Neural Networks

Greydanus et al. [6] introduced Hamiltonian Neural Networks (HNNs), demonstrating that learning a scalar Hamiltonian function $H(q, p)$ and deriving dynamics via Hamilton's equations $\dot{q} = \partial H/\partial p$, $\dot{p} = -\partial H/\partial q$ enables exact energy conservation by construction. The key insight is that the symplectic structure of Hamiltonian mechanics prevents the secular energy drift that afflicts standard neural ODEs, ensuring physically meaningful long-term trajectories. Subsequent work on symplectic integrators showed that the Störmer–Verlet leapfrog scheme preserves a shadow Hamiltonian $\tilde{H} = H + O(\varepsilon^2)$, bounding energy error uniformly for all time.

Our work applies HNNs to generative modeling in holographic boundary space, using the AdS gravitational potential as a structured prior within the learned Hamiltonian. The combination of a physics-motivated potential (pairwise Lorentz geodesic distances) with a learned correction (small MLP) ensures that the dynamics respect geometric structure while retaining data-adaptive capacity.

2.5 Bohmian Mechanics

David Bohm's pilot wave theory [1] provides a deterministic, realist interpretation of quantum mechanics in which particles follow definite trajectories guided by a wave function Ψ. The wave function evolves according to the Schrödinger equation and generates a velocity field $v = \nabla S/m$ where S is the phase of $\Psi = R\exp(iS/\hbar)$. The quantum potential $Q = -\hbar^2\nabla^2 R/(2mR)$ produces non-classical forces that account for interference and entanglement without wavefunction collapse.

The application of Bohmian guidance to multi-agent computation is, to our knowledge, novel. The closest analog is mean-field game theory, which uses PDE-based population dynamics but without the quantum potential structure that provides anti-bunching. In Avatar, the Hamiltonian ODE's momentum field serves as the pilot wave, and the quantum potential maintains diversity among Kuramoto clusters by preventing phase collapse.

2.6 Autopoiesis and Affective Neuroscience

Maturana and Varela [9] defined autopoiesis as the self-producing organization that distinguishes living from non-living systems: a living system continuously produces the components that constitute it. Friston's Free Energy Principle [5] formalized this as variational inference: an organism is a system that maintains its Markov blanket by minimizing variational free energy, and action and perception are the two mechanisms through which this minimization occurs.

Panksepp's affective neuroscience [11] identified seven primary emotional systems (SEEKING, RAGE, FEAR, LUST, CARE, PANIC, PLAY) as the foundation of mammalian consciousness, arguing that emotions are not cognitive epiphenomena but the primary motivational states that drive behavior. Damasio [4] complemented this with the somatic marker hypothesis: feelings—the subjective experience of bodily states—are essential for rational decision-making, and patients with damage to emotional brain circuits make poor decisions despite intact logical reasoning.

Our psyche layer draws from all four traditions: drives from Friston's FEP, emotions from Panksepp and Damasio, autopoietic self-production from Maturana and Varela, and narrative identity from the philosophy of personal identity.

2.7 Small Language Models for Embodied Cognition

The deployment of large language models as cognitive components in embodied agents has been explored extensively, but typically with models of 7B parameters or larger requiring dedicated GPU resources. The Qwen3 family [12] provides Apache 2.0-licensed models from 0.6B to 235B parameters, with the 0.6B variant (600M parameters) supporting both `/think` (chain-of-thought reasoning) and `/no_think` (fast reflexive) inference modes. This makes it uniquely suited for embodied cognitive architectures where the language model must share computational resources with a physics backbone: Qwen3 0.6B runs comfortably on CPU ($\sim$1.5 GB RAM) while the entire GPU remains available for the physics body. The Zamba2 architecture [15, 19] demonstrated that shared attention blocks with per-position LoRA adapters can match the performance of independent attention layers at a fraction of the parameter cost, a technique we adopt for the holographic attention in our backbone.

3 System Architecture Overview

The complete Avatar 4.0 system processes information through a three-layer hierarchy (Figure 1). At the bottom, the physics body converts raw sensory input (text embeddings from sentence-transformers `all-MiniLM-L6-v2`) into geometric representations on the Lorentz hyperboloid, processes them through a 60-layer reversible backbone with MERA-compressed feed-forward layers, evolves boundary coordinates via Hamiltonian dynamics with symplectic integration, and produces collective inference through Bohmian Kuramoto oscillators. The physics body outputs the Kuramoto order parameter $r \in [0, 1]$ measuring synchronization (pattern confidence) and the susceptibility χ (fluctuation-dissipation susceptibility: $\chi = N \cdot \mathrm{Var}(r)$, where r is the order parameter over a rolling window) measuring sensitivity to perturbation. Together, (r, χ) define the organism's position on the COP emotion manifold.

The psyche layer transforms these physics outputs into functional experience. Three drives (hunger, fatigue, curiosity) create functional needs; five emotions (satisfaction, pride, curiosity, boredom, anxiety) provide the primary decision mechanism; a self-model tracks competence and personality; and a circadian clock modulates arousal. The psyche's emotional state feeds back into the physics body by modulating Kuramoto coupling strength and drives the prefrontal cortex's query generation.

The prefrontal cortex (Qwen3 0.6B) provides cognitive interpretation of the organism's emotional and perceptual state. It generates search queries based on the current emotional context, interprets the meaning of detected patterns, and reflects on the organism's identity and development. Its output feeds back to the perception layer, closing the loop: the organism acts on its environment (web search), perceives the results, processes them through physics, computes an affect response via the COP manifold, and uses cognition to plan the next action.

Data flows through the system as follows. Each "tick" (default 60 seconds), the perception module embeds search results into a $(32, 2048)$ token tensor. The Lorentz embedding projects each token onto $\mathcal{H}^{64}$ with boundary coordinates $x \in \mathbb{R}^{64}$ and bulk depth $z \in \mathbb{R}^{+}$. The reversible backbone processes these tokens through 60 layers (50 SSM + 10 attention) with MERA-FFN, producing hidden representations. The Hamiltonian ODE initializes momenta from the backbone's output and integrates 3 leapfrog steps, producing final positions and momenta. The Bohmian Kuramoto module receives the momentum as pilot wave and evolves 32 phase clusters, outputting the order parameter r and per-cluster phase velocities. Actions are selected by the cluster with maximum phase velocity, modulated by emotional state. The psyche computes drives, maps (r, χ) to emotion via the COP manifold, updates the self-model, and passes context to the prefrontal cortex for query generation and interpretation.

4 Layer 1: The Physics Body

4.1 Lorentz Hyperboloid Embedding

Tokens are projected onto the Lorentz hyperboloid $\mathcal{H}^d = \{x \in \mathbb{R}^{d+1} : \langle x, x\rangle_L = -1/\kappa\}$ with $d = d_{\text{boundary}} = 64$ and learnable curvature $\kappa > 0$, following work on intrinsic Lorentz networks [8]. The Minkowski inner product is defined as:

$$\langle x, y\rangle_L = -x_0 y_0 + \sum_{i=1}^{d} x_i y_i \tag{1}$$

Given a spatial vector $x_{\text{spatial}} \in \mathbb{R}^d$ produced by a linear projection from the token embedding, the time component is computed as:

$$x_0 = \sqrt{\frac{1}{\kappa} + \|x_{\text{spatial}}\|^2} \tag{2}$$

which satisfies the hyperboloid constraint $\langle x, x\rangle_L = -x_0^2 + \|x_{\text{spatial}}\|^2 = -1/\kappa$ by construction. This eliminates the numerical instability of Poincaré ball models near the boundary ($\|x\| \to 1$) and provides a natural notion of "bulk depth" via a separate radial coordinate $z = \sigma(w^\top h + b)$ where h is the backbone hidden state.

The geodesic distance on the hyperboloid is:

$$d_{\text{geo}}(x, y) = \frac{1}{\sqrt{\kappa}} \operatorname{arccosh}(-\kappa\langle x, y\rangle_L) \tag{3}$$

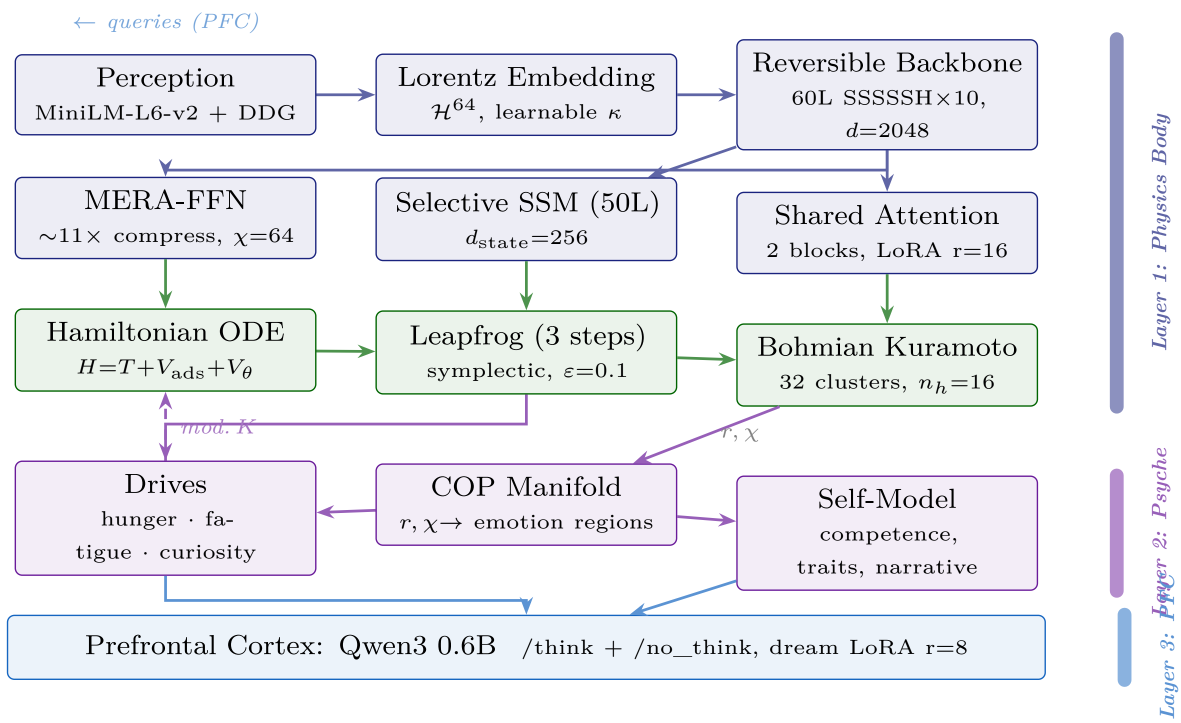

Figure 1: Three-layer architecture of Avatar 4.0. **Blue**: physics body processes perception through Lorentz embedding, MERA-compressed backbone, Hamiltonian dynamics, and Bohmian Kuramoto oscillators. **Purple**: psyche maps physics outputs (r, χ) via COP manifold to drives, emotions, and self-model. **Dark blue**: prefrontal cortex (Qwen3 0.6B) provides cognition and feeds queries back to perception. Solid arrows: forward data flow. Dashed arrows: feedback loops. Colored bars on right indicate layer boundaries.

The curvature κ is initialized to 1.0 and learned jointly with all other parameters. As training progresses, κ adapts to the intrinsic curvature of the data manifold: larger κ for highly hierarchical data (stronger curvature, more separation between scales) and smaller κ for flatter structures. This adaptive geometry is a key advantage over fixed-curvature embeddings.

4.2 MERA Tensor Train FFN

Each feed-forward layer in the backbone uses a MERA tensor train decomposition rather than a dense weight matrix. The standard SwiGLU FFN requires three weight matrices—gate, up, and down—each of size $d_{\text{model}} \times d_{\text{ff}}$ or its transpose. For $d_{\text{model}} = 2048$ and $d_{\text{ff}} = 4 \times d_{\text{model}}/3 \approx 2731$ (with SwiGLU correction), a single dense FFN layer contains approximately $3 \times 2048 \times 2731 \approx 16.8$M parameters.

The MERA decomposition replaces each dense matrix $W \in \mathbb{R}^{m\times n}$ with a chain of $k = n_{\text{cores}} = 4$ small core tensors connected by bond vectors of dimension $\chi = \text{bond\_dim} = 64$:

$$W \approx C_1 B_1 C_2 B_2 C_3 B_3 C_4 \tag{4}$$

where each core $C_i \in \mathbb{R}^{r_{i-1}\times d_i \times r_i}$ has bond dimensions $r_0 = m/k$, $r_k = n/k$, and $r_i = \chi = 64$ for interior bonds, and each bond vector $B_i \in \mathbb{R}^{\chi}$ mediates the connection between adjacent cores.

The three SwiGLU paths (gate, up, down) are each independently decomposed, preserving the activation pattern:

$$\text{FFN}(x) = W^{\text{MERA}}_{\text{down}}\Big(\text{SiLU}(W^{\text{MERA}}_{\text{gate}}x) \odot W^{\text{MERA}}_{\text{up}}x\Big) \tag{5}$$

Proposition 1 (Compression Ratio)**.** *The MERA-FFN achieves a compression ratio of:*

$$\begin{aligned}\rho &= \frac{3mn}{3k(r_{i-1}\cdot d_i \cdot r_i) + 3(k-1)\chi} \\ &= \frac{3\times 2048\times 2731}{3\times 4\times(512\times 64) + 9\times 64} \approx 11\end{aligned} \tag{6}$$

Across 60 layers, total FFN parameters drop from 60×16.8M $=$ 1,008M (if dense) to approximately 16M—a $\sim$11× compression. The remaining parameters are concentrated in the SSM ($d_{\text{state}} = 256$ per layer) and shared attention blocks (2 blocks with $n_{\text{heads}} = 16$, $d_{\text{head}} = 128$) where sequence modeling capacity is essential.

This compression is not merely a computational trick. MERA is the tensor network whose causal cone structure reproduces the Ryu–Takayanagi entanglement entropy formula [14]: $S_A = |\gamma_A| \log \chi$, where $|\gamma_A|$ is the number of bonds cut by the minimal surface γ_A. By using MERA for the FFN, we ensure that each layer's computation respects the same multi-scale geometric constraints as the AdS bulk, making the compression physically motivated rather than merely algebraically convenient.

4.3 Selective SSM (S7)

The 50 SSM layers implement selective state space models with input-dependent gating, following the S4/S5/S6/Mamba lineage [17]. Each layer maintains a latent state $h_t \in \mathbb{R}^{d_{\text{state}}}$ with $d_{\text{state}} = 256$ that evolves according to:

$$h_t = \bar{A} \cdot h_{t-1} + \bar{B} \cdot x_t \tag{7}$$

$$y_t = C \cdot h_t + D \cdot x_t \tag{8}$$

where the discretized matrices $\bar{A}$ and $\bar{B}$ are computed from continuous parameters via the zero-order hold:

$$\bar{A} = \exp(\Delta \cdot A) \tag{9}$$

$$\bar{B} = (\Delta \cdot A)^{-1}(\exp(\Delta \cdot A) - I) \cdot \Delta \cdot B \tag{10}$$

The *selective* aspect is that Δ, B, and C are functions of the input x_t, computed by linear projections:

$$\Delta_t = \text{softplus}(W_\Delta x_t + b_\Delta), \quad B_t = W_B x_t, \quad C_t = W_C x_t \tag{11}$$

This input-dependent gating allows each SSM layer to selectively attend to or ignore input tokens based on content, providing a mechanism analogous to attention's content-based routing but with linear computational complexity in sequence length. The parallel scan algorithm computes the full sequence in $O(N \log N)$ time on GPU, enabling efficient training.

4.4 Shared Holographic Attention

At positions 5, 11, 17, 23, 29, 35, 41, 47, 53, and 59 in the 60-layer stack (every 6th layer), the SSM is replaced by holographic attention. Following Zamba2 [15], only 2 attention blocks are instantiated, and each of the 10 positions routes to one of these 2 blocks. Each block has $n_{\text{heads}} = 16$ heads with $d_{\text{head}} = 128$, and each position maintains its own LoRA adapter (rank 16) applied to the query and value projections:

$$W_q^{(i)} = W_q^{\text{shared}} + \alpha \cdot A_q^{(i)} B_q^{(i)}, \quad W_v^{(i)} = W_v^{\text{shared}} + \alpha \cdot A_v^{(i)} B_v^{(i)} \tag{12}$$

where $A^{(i)} \in \mathbb{R}^{d_{\text{model}} \times 16}$ and $B^{(i)} \in \mathbb{R}^{16 \times d_{\text{model}}}$ are the per-position LoRA factors, with 5 adapters per shared block (one per pair of positions).

The attention kernel incorporates AdS geometry through the geodesic distance:

$$K_{ij}^{(\Delta)} = \left(\frac{1}{\cosh(d_{\text{geo}}(x_i, x_j)) + 1} \right)^{\exp(\log \delta_h)} \tag{13}$$

where δ_h is a learnable per-head parameter controlling the sharpness of geometric attention. This kernel implements the AdS bulk-to-boundary propagator: tokens that are close in hyperbolic space (small geodesic distance) attend strongly to each other, while distant tokens interact weakly. The exponential parameterization $\exp(\log \delta_h)$ ensures positivity.

4.5 Reversible Residual Coupling

The backbone splits the residual stream into two halves (h_1, h_2) of $d_{\text{model}}/2 = 1024$ each. Each layer updates one half from the other via additive coupling:

$$h_1^{(\ell+1)} = h_1^{(\ell)} + F^{(\ell)}(h_2^{(\ell)}) \tag{14}$$

$$h_2^{(\ell+1)} = h_2^{(\ell)} + G^{(\ell)}(h_1^{(\ell+1)}) \tag{15}$$

where $F^{(\ell)}$ is the core layer (SSM or attention) and $G^{(\ell)}$ is the MERA-FFN at layer ℓ. During backpropagation, activations are exactly reconstructed by running the reverse equations:

$$h_2^{(\ell)} = h_2^{(\ell+1)} - G^{(\ell)}(h_1^{(\ell+1)}) \tag{16}$$

$$h_1^{(\ell)} = h_1^{(\ell+1)} - F^{(\ell)}(h_2^{(\ell)}) \tag{17}$$

This yields **zero activation memory** regardless of network depth. Standard backpropagation through 60 layers would require storing all intermediate activations, consuming $O(L \cdot d_{\text{model}} \cdot N)$ memory. With reversible coupling, only the final-layer activations need to be stored, and all previous activations are recomputed on the fly during the backward pass. This is essential for fitting the full forward+backward pass into ∼4.6 GB VRAM—without reversibility, the activation memory alone would exceed the GPU's capacity.

4.6 Page Curve Memory

The backbone maintains a ring buffer implementing holographic information bounds inspired by the Page curve from black hole physics. The Page curve describes how the entanglement entropy of Hawking radiation initially increases as a black hole evaporates, reaches a maximum at the Page time (when the radiation's Hilbert space dimension equals the black hole's), and then decreases as the black hole becomes maximally entangled with its radiation.

In Avatar, the ring buffer stores up to max_cache = 128 hidden states. When the buffer is full, new tokens evict old tokens via an "island" mechanism: the island_size = 32 oldest entries form an "island" that is compressed into a single summary vector and evicted. The Bekenstein bound enforces an information capacity limit:

$$S_{\text{Bek}} = \frac{2\pi RE}{\hbar c} \leq S_{\max} = d_{\text{boundary}} \cdot \log 2 \tag{18}$$

where R is the effective radius of the boundary region (computed from the spatial extent of stored embeddings) and E is the total energy (sum of squared norms). When the stored entropy exceeds the Bekenstein bound, islands are forcibly evicted regardless of age, maintaining the holographic principle's requirement that boundary information content cannot exceed the area of the enclosing surface.

A regularization loss $\mathcal{L}_{\text{page}} = \alpha \cdot \max(0, S - S_{\max})$ softly enforces this bound during training.

5 Hamiltonian Dynamics

5.1 Learned Hamiltonian

The generative dynamics operate in the $d_b = d_{\text{boundary}} = 64$-dimensional Lorentz boundary space via a phase space (q, p) where $q \in \mathbb{R}^{N \times d_b}$ represents boundary positions of N tokens and $p \in \mathbb{R}^{N \times d_b}$ represents conjugate momenta. The Hamiltonian decomposes into kinetic energy, AdS gravitational potential, and a learned correction:

$$H(q, p) = \underbrace{\frac{1}{2}\|p\|^2}_{T(p)} + \underbrace{V_{\text{ads}}(q)}_{\textit{gravitational}} + \underbrace{V_\theta(q)}_{\textit{learned}} \tag{19}$$

Hamilton's equations of motion are:

$$\dot{q}_i = \frac{\partial H}{\partial p_i} = p_i, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i} = -\nabla_{q_i} V_{\text{ads}} - \nabla_{q_i} V_\theta \tag{20}$$

The kinetic energy $T(p) = \frac{1}{2}\sum_i \|p_i\|^2$ is the standard Euclidean norm, giving $\partial T/\partial p_i = p_i$. The separation $H = T(p) + V(q)$ is essential for the leapfrog integrator, which requires that the kinetic and potential terms depend on different phase-space variables.

5.2 AdS Gravitational Potential

The AdS gravitational potential creates attractive wells between token positions using the Lorentz geodesic distance:

$$V_{\text{ads}}(q) = \sum_{i<j} \log(\cosh(d_{\text{geo}}(q_i, q_j)) + \epsilon) \tag{21}$$

where d_{geo} is the Lorentz geodesic distance (Eq. 3) and $\epsilon = 10^{-6}$ prevents logarithmic singularity when tokens coincide. This potential has several desirable properties: it is attractive at long range (pulling distant tokens toward each other, encouraging clustering), approximately flat at short range (the cosh saturates near zero distance, preventing collapse), and respects the hyperbolic geometry (distances are measured along geodesics of the hyperboloid, not in ambient Euclidean space).

The gradient of V_{ads} with respect to q_i is:

$$\nabla_{q_i} V_{\text{ads}} = \sum_{j \neq i} \frac{\sinh(d_{\text{geo}}(q_i, q_j))}{\cosh(d_{\text{geo}}(q_i, q_j)) + \epsilon} \cdot \nabla_{q_i} d_{\text{geo}}(q_i, q_j) \tag{22}$$

where $\nabla_{q_i} d_{\text{geo}}$ is computed via automatic differentiation through the Lorentz inner product and arccosh.

The learned potential V_θ is a small MLP ($d_b \to 64 \to 1$, approximately 6K parameters) that captures data-specific corrections to the gravitational prior. By keeping the learned component small relative to the structured prior, we bias the dynamics toward physically meaningful trajectories while retaining data-adaptive capacity.

5.3 Leapfrog Symplectic Integration

Hamilton's equations are integrated via the Störmer–Verlet (leapfrog) scheme with $n = 3$ steps and step size $\varepsilon = 0.1$. Each step consists of a half-step momentum update, a full position update, and another half-step momentum update:

$$p_{t+1/2} = p_t - \frac{\varepsilon}{2} \nabla_q V(q_t) \tag{23}$$

$$q_{t+1} = q_t + \varepsilon \cdot p_{t+1/2} \tag{24}$$

$$p_{t+1} = p_{t+1/2} - \frac{\varepsilon}{2} \nabla_q V(q_{t+1}) \tag{25}$$

Proposition 2 (Energy Conservation)**.** *The leapfrog integrator is a second-order symplectic integrator that preserves the symplectic two-form $\omega = \sum_i dq_i \wedge dp_i$ exactly. It conserves a shadow Hamiltonian $\tilde{H} = H + O(\varepsilon^2)$, so the energy error $|H(q_T, p_T) - H(q_0, p_0)| = O(\varepsilon^2)$ per step with* ***no secular drift****, guaranteeing bounded trajectories for all time.*

This is the key advantage over standard neural ODEs or flow matching: the leapfrog integrator cannot produce runaway trajectories. Regardless of what the learned potential V_θ predicts, the symplectic structure ensures that phase-space volume is preserved and energy remains bounded. With $n = 3$ steps and $\varepsilon = 0.1$, the total integration time is $T = 0.3$, sufficient for the system to explore local neighborhoods of the energy landscape without drifting far from the initial configuration.

5.4 Momentum Initialization

The initial momenta p_0 are not drawn from a random distribution (as in Hamiltonian Monte Carlo) but are *conditioned on the backbone's output*. Specifically, the backbone's final hidden state $h_{\text{out}} \in \mathbb{R}^{N\times d_{\text{model}}}$ is projected to boundary dimension:

$$p_0 = W_p \cdot h_{\text{out}} + b_p, \quad W_p \in \mathbb{R}^{64\times 2048} \tag{26}$$

This conditioning is critical: it allows the backbone's learned representations to influence the direction of Hamiltonian evolution. The backbone encodes information about token relationships, semantic content, and geometric structure; by projecting this into momentum space, we ensure that the Hamiltonian dynamics explore directions relevant to the data rather than random phase-space regions. The evolved momenta p_T after leapfrog integration then serve as the pilot wave for the Bohmian Kuramoto oscillators, completing the chain from backbone representation through Hamiltonian physics to collective inference.

5.5 Training Loss

The training loss combines three terms:

$$\mathcal{L} = \underbrace{\|q_T - q_{\text{data}}\|^2}_{\mathcal{L}_{\text{recon}}} + \underbrace{\lambda_E(H_T - H_0)^2}_{\mathcal{L}_{\text{energy}}} + \underbrace{\lambda_S(-\bar{r})}_{\mathcal{L}_{\text{sync}}} \tag{27}$$

The reconstruction loss $\mathcal{L}_{\text{recon}}$ ensures that the evolved positions match the target data. The energy conservation loss $\mathcal{L}_{\text{energy}}$ penalizes violations of the Hamiltonian constraint, encouraging the learned potential V_θ to produce dynamics compatible with symplectic integration. The synchronization loss $\mathcal{L}_{\text{sync}}$ maximizes the mean Kuramoto order parameter $\bar{r}$, encouraging the oscillator ensemble to find coherent patterns in the data.

Optimization uses Adafactor [13] with factored second moments, which reduces optimizer state memory from approximately ∼0.85 GB (Adam with two moment buffers for 106.2M parameters) to approximately 2 MB by factoring the second-moment matrix into row and column statistics. This reduction is essential for fitting forward+backward within the 6 GB VRAM budget of the GTX 1660 Ti.

6 Predictive Processing: Per-Tick Body Learning

The organism does not merely process sensory inputs—it *predicts* what it will perceive and learns continuously from the difference between prediction and reality. This section describes the per-tick online learning mechanism enabled by the $d_{\text{model}} = 2048$ architecture choice.

6.1 Prediction Error and the Learning Heartbeat

Every tick, the organism performs both a forward pass *and* a backward pass through the physics backbone. The prediction error is defined as:

$$\varepsilon = \|q_{\text{final}} - q_{\text{data}}\|^2 \tag{28}$$

where q_{final} is the Hamiltonian-evolved boundary position and q_{data} is the target embedding from perception. The backward pass computes gradients with respect to all MERA cores and the learned Hamiltonian potential V_θ, and a gradient step updates these weights immediately.

The physics body's weights therefore *change every heartbeat*. This is not dreaming or offline consolidation—it is the body adapting to every new experience in real time, exactly as a biological nervous system does. Formally, each tick executes:

$$q_T, p_T = \text{LeapFrog}(q_0, p_0; H) \tag{29}$$

$$\varepsilon = \|q_T - q_{\text{data}}\|^2 \tag{30}$$

$$\theta \leftarrow \theta - \eta\nabla_\theta\varepsilon \tag{31}$$

6.2 Why d_{model} Was Reduced from 3072 to 2048

At $d_{\text{model}} = 3072$ the forward pass occupies 4.8 GB VRAM—leaving no room for the gradient buffers required by backpropagation (>6 GB combined, causing OOM). The organism could run with a frozen body, but a frozen body cannot learn from experience.

By reducing d_{model} to 2048, the forward pass drops to ∼3.5 GB and the full forward+backward cycle fits in 5,460 MiB—within the 6 GB budget of the GTX 1660 Ti. A smaller model that learns continuously is fundamentally more alive than a larger model that is frozen. The ∼11× MERA compression ensures that the 106.2M-parameter body retains sufficient representational capacity despite the reduced width.

6.3 Empirical Impact of Per-Tick Learning

Table 1 compares the organism's synchronization trajectory under a frozen body ($d_{\text{model}} = 3072$) versus per-tick learning ($d_{\text{model}} = 2048$).

Table 1: Per-tick learning (2048) vs. frozen body (3072). With per-tick learning the organism reaches near-full synchronization in 4 ticks; the frozen body oscillates in boredom/anxiety for 85 ticks.

Tick	r	Emotion	Notes
Per-tick learning ($d_{\text{model}} = 2048$, forward+backward every tick)			
1	0.708	Pride	Body already synchronizing
2	0.943	Pride	Rapid improvement
3	0.961	Pride	Near-full sync
4	0.969	Satisfaction	Prediction error dropping
Frozen body ($d_{\text{model}} = 3072$, no per-tick gradient steps)			
1–85	0.15–0.22	Boredom/Anxiety	Stuck; cannot adapt

The contrast is stark. With a frozen body, the organism oscillated between boredom and anxiety at $r = 0.15$–0.22 for 85 ticks, unable to adapt its internal representations to the incoming data. With per-tick learning, the organism reached $r = 0.969$ (satisfaction) in just 4 ticks, demonstrating that continuous online learning is not a luxury but the mechanism that makes the organism *responsive* rather than merely reactive.

7 Black-Scholes Volatility Surface

The organism must continuously decide *what to explore next*—a multi-armed bandit problem with non-stationary rewards. We reformulate this as option pricing: each candidate research topic is a call option whose value reflects the expected probability of discovery.

7.1 Topic as Call Option

Define the option parameters for topic τ:

$$S_\tau = \mathrm{EMA}_\alpha(r_\tau) \quad \text{(spot price: current competence)} \tag{32}$$

$$K = 0.6 \quad \text{(strike: discovery threshold)} \tag{33}$$

$$\sigma_\tau = \mathrm{clip}(10 \cdot \mathrm{std}(|\Delta \mathrm{FE}_\tau|),\ 0.05,\ 2.0) \tag{34}$$

$$T = \frac{\text{ticks until dream}}{\text{dream interval}} \in [0, 1] \tag{35}$$

where the EMA uses $\alpha = 0.3$ over a rolling window of 20 observations per topic. The implied volatility σ_τ measures how unpredictable the organism's prediction errors are on topic τ: high σ means the topic still surprises the organism, indicating unexplored structure.

The option value is computed via the standard Black-Scholes call formula [24]:

$$d_1 = \frac{\ln(S/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{36}$$

$$V_\tau = S \cdot \Phi(d_1) - K \cdot \Phi(d_2) \tag{37}$$

where Φ is the standard normal CDF (risk-free rate $r = 0$; no discounting in a biological organism).

7.2 Exploration Value Curve

The Black-Scholes formula naturally encodes the explore-exploit tradeoff:

- **Low S, low σ** (deep OTM, low vol): worthless—the organism cannot reach discovery and the topic doesn't surprise it.
- **Low S, high σ** (OTM, high vol): moderate value—volatile topics might produce unexpected breakthroughs.
- **$S \approx K$, any σ** (at the money): *highest value*—the organism is at the edge of discovery.
- **High S, high σ** (ITM, high vol): good—mastered but still surprising.
- **High S, low σ** (deep ITM, low vol): *low value*—mastered and predictable ("priced in").

7.3 Novelty Discount for Mastered Topics

Deep-ITM topics with low volatility represent mastery without novelty. We apply a satiation discount:

$$V_\tau^{\text{eff}} = \begin{cases} V_\tau \cdot \left(1 - \min\left(1, \frac{S_\tau - K}{0.3}\right)\right) + (V_\tau - \max(0, S_\tau - K)) \\ V_\tau \end{cases} \tag{38}$$

This ensures that mastered, predictable topics are deprioritized even though their intrinsic value is positive, while mastered topics that remain volatile (still producing surprises) retain their option value.

7.4 Integration with Query Decision

The volatility surface is consulted at two points in the organism's 5-layer query decision cascade:

1. **Emergency overrides** (starvation, frustration): Instead of selecting a random seed topic, the organism selects the topic with highest V_τ^{eff}—even in crisis, it makes the best available choice.
2. **Satiation/boredom switching** (Layer 4): The organism switches topic only if the best alternative has $\geq 20\%$ higher option value than the current topic, preventing restless oscillation.

Unknown topics (fewer than 3 observations) receive a default implied volatility of $\sigma = 1.0$—an optimistic exploration prior that ensures unexplored territory competes with familiar topics on equal footing.

7.5 Why Not Stochastic Hamiltonian?

A natural extension would inject σ-scaled Brownian noise into the leapfrog integrator, converting the deterministic Hamiltonian ODE into a stochastic differential equation. However, naive noise injection destroys the symplectic structure that guarantees energy conservation. A proper implementation requires a stochastic symplectic integrator (e.g., splitting methods for Langevin dynamics), which we defer to future work.

8 Dream Replay: Biological Five-Phase Sleep

While per-tick learning adapts the physics body to individual experiences, the nightly dream cycle consolidates, recombines, and extends learning across the organism's entire episodic history. The dream cycle implements a five-phase biological protocol inspired by mammalian sleep architecture: (1) body replay via CLion on GPU subprocess, (2) FineWeb active learning on GPU subprocess, (3) dream visitors (Whisper+Kokoro on CPU, then GPU training), (4) mind LoRA fine-tuning on CPU, and (5) GEPA prompt evolution on CPU+Ollama.

8.1 Phase 1: Slow-Wave Replay

The organism replays real episodes from the SQLite store, re-processing high-confidence memories ($r > 0.6$) through the backbone.

- **Duration:** 10 gradient steps
- **Gradient scale:** 10% of standard learning rate
- **Purpose:** Strengthen existing representations; reduce prediction error on confirmed discoveries

The reduced gradient scale prevents catastrophic interference: the replay reinforces successful patterns without overwriting the representations formed during waking per-tick learning.

8.2 Phase 2: REM Recombination

The organism mixes token sequences from *different* episodes to form novel associative connections—the computational equivalent of REM sleep's role in creative insight.

- **Duration:** 5 gradient steps
- **Gradient scale:** 5% of standard learning rate
- **Purpose:** Build cross-domain associations; enable metaphorical reasoning across topics

Token mixing is performed by sampling pairs of episodes from distinct topic clusters and concatenating their token sequences. The MERA cores learn to represent the structural similarities between superficially different domains, enabling the PFC to generate cross-disciplinary queries during the next waking cycle.

8.3 Phase 3: Counterfactual Imagination

The organism generates counterfactual trajectories by adding structured noise to real episode embeddings and integrating forward through the Hamiltonian ODE.

- **Duration:** 5 gradient steps
- **Gradient scale:** 2% of standard learning rate
- **Purpose:** Explore nearby phase-space neighborhoods; develop robustness to perturbation

Gradient descent on these imagined trajectories pushes the backbone toward representations that remain stable under small perturbations of sensory input, improving generalization.

8.4 The Body Grows During Sleep

Phase 1 (body replay) updates the backbone weights directly—not just the LoRA adapters in the prefrontal cortex. Combined with per-tick learning during waking hours, this means the physics body is never frozen: it adapts at two timescales, tick-by-tick during waking (fast adaptation) and nightly during dreaming (slow consolidation).

8.5 CLion: Memory-Efficient Dream Optimizer

The body dream uses CLion (Cautious Lion) [22], a sign-based optimizer that solves two simultaneous problems on the 6 GB GPU. Adam requires two moment buffers consuming ∼849 MB on top of the model's ∼3.5 GB forward footprint—exceeding the budget. CLion maintains only one momentum buffer (∼488 MB) via the update rule:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \tag{39}$$

$$u_t = \text{sign}(m_t) \cdot \mathbf{1}[g_t \cdot \text{sign}(m_t) > 0] \tag{40}$$

The binary cautious mask $\mathbf{1}[\cdot]$ ensures no coordinate update ever increases the loss, preventing the runaway divergence that produces NaN weights when Lorentz boundary coordinates drift outside the hyperboloid after extended operation. Combined with gradient clipping ($\|\nabla\|_2 \leq 0.1$) before the sign operation, CLion makes the body dream robust to numerical instabilities that accumulate over hundreds of waking ticks.

8.6 Sequential Execution: Five-Phase Dream Pipeline

A critical implementation detail: the five dream phases must run *sequentially*, never simultaneously. On a 6 GB GPU with 16 GB RAM, running JAX gradient computation alongside PyTorch model loading causes fatal out-of-memory crashes (exit code 137). The solution is a five-phase sequential pipeline:

1. **Phase 1: Body replay (CLion, GPU subprocess).** Save physics checkpoint to disk. Run CLion body replay (10 replay + 5 recombine + 5 imagine gradient steps). Save updated checkpoint. Subprocess exits, reclaiming all GPU memory.
2. **Phase 2: FineWeb active learning (GPU subprocess).** A second subprocess loads the updated checkpoint and trains on FineWeb-Edu batches selected by the active sampler. Subprocess exits cleanly.

3. **Phase 3: Dream visitors (Whisper+Kokoro CPU, then GPU training).** Whisper (39M, int8) transcribes archived audio; Kokoro (82M, ONNX) synthesises speech from discoveries. A GPU subprocess trains the AudioFNO on enriched (audio, text) pairs via InfoNCE contrastive alignment. Subprocess exits.
4. **Phase 4: Mind (LoRA fine-tuning, CPU).** Load Qwen3 0.6B via `transformers`, apply LoRA (rank 8), fine-tune for ∼50 steps on the organism's episode history (∼1.5 GB RAM). Save adapter weights to `data/pfc_adapter/`.
5. **Phase 5: GEPA prompt evolution (CPU+Ollama).** Avatar's episode store is queried for high-r and low-r trajectories. Qwen3 0.6B (via Ollama) reflects on the contrast and proposes improved query and reflection instructions, saved to `data/pfc_prompts.json` (∼3 min, ∼0 extra RAM).

After all five phases, the organism reloads the updated physics checkpoint onto GPU. The prefrontal cortex loads `base+adapter` plus evolved prompt instructions on the next inference call.

This sequential protocol ensures that all five phases undergo real weight modification during each dream cycle, without memory contention. After dreaming, the organism wakes up with improved physics weights (Phase 1 CLion body dream), corpus-trained representations (Phase 2 FineWeb), matured sensory codes (Phase 3 dream visitors), personalized language model weights (Phase 4 LoRA mind dream), and evolved prompt instructions (Phase 5 GEPA)—five independent mechanisms of experiential growth operating on different timescales.

9 Bohmian Pilot Wave Dynamics

9.1 Standard Kuramoto Model

The classical Kuramoto model describes the synchronization of K coupled oscillators with phases $\theta_k \in [0, 2\pi)$ and natural frequencies ω_k:

$$\frac{d\theta_k}{dt} = \omega_k + \frac{\kappa}{K}\sum_{j=1}^{K}\sin(\theta_j - \theta_k) \tag{41}$$

where κ is the coupling strength. The mean-field order parameter $re^{i\psi} = K^{-1}\sum_j e^{i\theta_j}$ measures global synchronization: $r = 0$ indicates incoherence, $r = 1$ indicates perfect phase locking. Above a critical coupling $\kappa_c = 2/(\pi g(0))$ (where g is the frequency distribution), a macroscopic fraction of oscillators spontaneously synchronize—a phase transition that has been used to model neural synchronization, power grid stability, and collective decision-making.

In Avatar, we extend the Kuramoto model to $n_h = 16$-dimensional phases, with $K = 32$ clusters. Each cluster represents a group of correlated features, and synchronization across clusters indicates pattern detection in the input data.

9.2 Bohmian Extension: Pilot Wave from Momentum Field

We replace the mean-field coupling term with Bohmian pilot wave guidance:

Definition 1 (Bohmian Kuramoto Dynamics)**.**

$$\frac{d\theta_k}{dt} = \omega_k + \underbrace{\kappa \cdot p_{\text{final}}^{(k)}}_{\nabla S\ (\textit{pilot wave})} + \underbrace{Q(\theta_1, \ldots, \theta_K)}_{\textit{quantum potential}} + \underbrace{\eta_k \cdot \text{obs}_k}_{\textit{observation}} \tag{42}$$

The pilot wave $p_{\text{final}}^{(k)}$ is the Hamiltonian ODE's evolved momentum, projected to cluster space via the observation bridge's assignment matrix $M \in \mathbb{R}^{K\times N}$:

$$p_{\text{final}}^{(k)} = \sum_{i=1}^{N} M_{ki} \cdot p_T^{(i)} \tag{43}$$

This connects the swarm's behavior to the backbone's learned physics: the oscillators are guided by the same energy landscape that drives the generative dynamics. The pilot wave provides directed, coherent guidance (analogous to Bohm's ∇S term), while the quantum potential (Section 9.3) provides the non-local correlations that prevent degenerate synchronization. The observation term $\eta_k \cdot \text{obs}_k$ couples each cluster to its assigned sensory observations, grounding the dynamics in external data.

9.3 Quantum Potential

The quantum potential provides a non-local antibunching force that maintains diversity among oscillator clusters:

$$Q_k = -\left(\log|\psi_k| - \overline{\log|\psi|}\right) \cdot (\theta_k - \bar{\theta}) \tag{44}$$

where the effective wavefunction amplitude is:

$$|\psi_k| \propto \exp\left(-\frac{1}{2}\|\theta_k - \bar{\theta}\|^2\right) \tag{45}$$

and $\bar{\theta} = K^{-1}\sum_j \theta_j$ is the mean phase.

When clusters converge in phase space ($\theta_k \approx \bar{\theta}$), $\log|\psi_k|$ is large and Q_k pushes them apart (antibunching). When they spread too far ($\|\theta_k - \bar{\theta}\|$ large), $\log|\psi_k|$ is small (below the mean) and Q_k pulls them toward coherence. This implements the quantum-mechanical principle of anti-bunching: identical bosonic particles spread out to maximize their phase-space coverage, preventing collapse into a single mode.

In the context of collective inference, the quantum potential solves the explore-exploit dilemma without explicit entropy regularization. Clusters that detect genuine patterns synchronize (exploit); clusters in uninformative regions are pushed apart by Q to explore new phase-space neighborhoods. The balance is automatic and adaptive, determined by the local density of the phase distribution.

9.4 Order Parameter and Synchronization Measure

The order parameter for the n_h-dimensional Kuramoto model is computed as a generalized Kuramoto parameter:

$$r = \left\| \frac{1}{K} \sum_{k=1}^{K} \frac{\theta_k}{\|\theta_k\|} \right\| \tag{46}$$

This reduces to the standard Kuramoto r when $n_h = 1$. Values of $r > 0.6$ indicate strong synchronization (pattern detected); $r < 0.4$ indicates incoherence (no pattern); and $r \approx 0.5$ indicates partial synchronization (emerging pattern, not yet resolved).

The order parameter is the primary interface between the physics body and the psyche layer. It encodes pattern confidence in a single scalar that, together with the susceptibility χ, determines the organism's position on the COP emotion manifold: high r with low χ produces satisfaction, high r with high χ produces pride, while low r maps to boredom (low χ) or anxiety (high χ).

9.5 Action Selection from Phase Velocity

Actions are selected based on the phase velocity of each cluster after Bohmian evolution. The per-cluster phase velocity is:

$$v_k = \left\| \frac{d\theta_k}{dt} \right\| = \left\| \omega_k + \kappa \cdot p_{\text{final}}^{(k)} + Q_k + \eta_k \cdot \text{obs}_k \right\| \tag{47}$$

The cluster with maximum phase velocity determines the primary action direction, while the velocity distribution across clusters determines action confidence. A concentrated distribution (one dominant cluster) produces decisive action; a flat distribution (many clusters with similar velocity) produces exploratory, diffuse action. This action selection mechanism is modulated by the psyche's emotional state: when anxious, only the highest-velocity cluster is considered (narrow, risk-averse behavior); when curious, a broader distribution of clusters contributes (wide, exploratory behavior).

9.6 Bohm's Holomovement Mapping

The architecture realizes David Bohm's ontological framework [1] not as metaphor but as structural analogy with precise computational counterparts. Table 2 provides the detailed correspondence.

Table 2: Bohm's holomovement mapped to Avatar 4.0.

Bohm Concept	**Avatar Implementation**
Implicate Order	MERA bulk: info enfolded via hierarchical tensor contractions, $\chi = 64$
Holomovement	Hamiltonian ODE: bulk→boundary unfolding via symplectic leapfrog
Explicate Order	Lorentz boundary tokens ($\mathcal{H}^{64}$): outputs, actions, perceptions
Pilot Wave ∇S	Momentum p_{final}: phase gradient conditioned on backbone states
Quantum Potential	Anti-bunching force (Eq. 44): diversity via non-local correlations
Active Information	Coupling $\eta_k \cdot \text{obs}_k$: sensory data *informs* trajectory
Soma-significance	Psyche: (r, χ) acquires emotional significance via COP manifold, feeds back to body
Undivided Wholeness	Closed loop: all layers coupled; none exists independently

The key insight is that Bohm's holomovement is not just a philosophical position but a computational architecture: information enfolds into bulk tensors (implicate), unfolds through Hamiltonian dynamics (holomovement), manifests as boundary representations (explicate), and is guided by momentum fields (pilot wave) with non-local correlations (quantum potential). Avatar makes this computationally concrete.

10 Layer 2: The Psyche

A physics engine processes inputs. An organism *lives*. The psyche layer transforms Avatar from a computational pipeline into an autopoietic agent [9] with functional needs, physics-grounded affect [4], and narrative identity. Every component of the psyche is grounded in the physics body's outputs—there is no separate "mind stuff," only organized matter whose dynamics constitute its affective states.

10.1 Allostatic Drives

Three primary drives create biological-like necessities that, if unmet, degrade the organism's functioning. These drives implement the Free Energy Principle's core claim [5]: organisms act to minimize surprise because their continued existence depends on it. Critically, these are not loss terms to be optimized—they are *functional states that modulate behaviour and are resolved through action.*

Information Hunger $h \in [0,1]$**.** Hunger increases when the system has not reduced free energy recently:

$$h_{t+1} = \text{clip}(h_t + \alpha_h - \beta_h \cdot \max(0, -\Delta\text{FE}_t),\ 0,\ 1) \tag{48}$$

where α_h is the baseline hunger rate and β_h scales the satiation from learning. When hungry ($h > 0.7$), the organism searches more aggressively. Starvation ($h > 0.85$) triggers desperate broadening of search scope. Satiation ($h < 0.3$) after successful learning (negative ΔFE) allows the organism to rest and consolidate.

Fatigue $f \in [0,1]$**.** Fatigue accumulates continuously during waking at a rate modulated by hunger:

$$f_{t+1} = \text{clip}(f_t + \alpha_f \cdot (1 + h_t),\ 0,\ 1) \tag{49}$$

Hungry organisms tire faster. When fatigue exceeds 0.8, processing quality degrades (the Kuramoto coupling strength κ is reduced, simulating cognitive impairment) and dream pressure becomes critical. Only dreaming resets fatigue. This creates functional *need* for sleep—not a scheduled maintenance window but a drive-level urgency.

Curiosity $c \in [0,1]$**.** Unlike hunger (which drives general information seeking), curiosity is specifically attracted to *novelty at the edge of understanding*—topics where the Kuramoto order parameter $r \approx 0.5$:

$$c = \exp\left(-\frac{1}{2}\left(\frac{r - 0.5}{0.15}\right)^2\right) \tag{50}$$

This Gaussian peaks exactly where the organism has enough understanding to form hypotheses ($r > 0$) but not enough to confirm them ($r < 1$). Maximum curiosity occurs at the boundary between order and chaos—the "edge of understanding" where discovery is most likely.

Satiation. When per-tick body learning is active, the Kuramoto oscillators can collapse to $r = 1.0$ (permanent synchronization), trapping the organism in perpetual pride with no exploration. Satiation prevents this by creating restlessness after sustained high synchronization. After N consecutive ticks with $r > 0.7$, satiation s accumulates:

$$s_{t+1} = \begin{cases} \min(1,\ s_t + 0.08) & \text{if } r > 0.7 \\ \max(0,\ s_t - 0.1) & \text{otherwise} \end{cases} \tag{51}$$

When $s > 0.6$, the organism actively reduces Kuramoto coupling K by up to 80%, forcing desynchronization. Satiation also boosts curiosity: $c_{\text{eff}} = \min(1,\ c_{\text{base}} + 0.8\,s)$, ensuring the organism seeks novelty even when the current topic is well-understood. This produces the biological explore-exploit cycle: discover → synchronize → pride → satiation → desynchronize → curiosity → explore → discover. In live deployment, this cycle completes in approximately 12–15 ticks.

10.2 Emotional Valence Space: The COP Emotion Manifold

The *Critical Order Parameter* (COP) engine maps the Kuramoto order parameter r and susceptibility χ (fluctuation-dissipation susceptibility: $\chi = N \cdot \text{Var}(r)$, where r is the order parameter over a rolling window) onto a continuous manifold. Emotions are *regions* of the (r, χ) plane, and the SOC controller drives the coupling K toward the critical line $K = K_c$ where χ diverges and the system is maximally open to reconfiguration.

This mirrors Damasio's somatic marker hypothesis [4]: emotions are not obstacles to rational decision-making but its essential foundation. When bored (low r, low χ), the organism jumps to its least-explored area. When anxious (low r, high χ), it retreats to topics where its self-model records high historical competence. When curious (mid r, high χ), it lingers at the critical edge, exploring adjacent hypotheses. When proud (high r, high χ), it digs deeper into its discovery, seeking to extend the pattern. When satisfied (high r, low χ), it consolidates, recording the finding in narrative memory and updating competence.

10.3 COP Equations

The three macroscopic observables are computed each tick:

Susceptibility. Following the fluctuation-dissipation theorem:

$$\chi = N \cdot \text{Var}_w(r), \quad \chi_{\text{norm}} = \frac{\chi}{\max_{t' \le t} \chi(t')} \tag{52}$$

where Var_w denotes variance over a rolling window of $w = 20$ ticks and $N = K \times n_h = 512$.

Relaxation time. From the autocorrelation of r:

$$\tau = \frac{1}{w-1} \sum_{i=1}^{w-1} \frac{(r_i - \bar{r})(r_{i-1} - \bar{r})}{\text{Var}(r) + \epsilon} \tag{53}$$

SOC controller. A proportional controller drives coupling K toward the critical point:

$$\dot{K} = \eta \cdot (0.5 - r) \cdot \max(\chi, 0.1), \quad K \in [0.05, 2.0], \quad \eta = 0.005 \tag{54}$$

Unity index. Eigenvalue dominance of the time-averaged coherence matrix:

$$U = \frac{\lambda_1}{\sum_k \lambda_k} \tag{55}$$

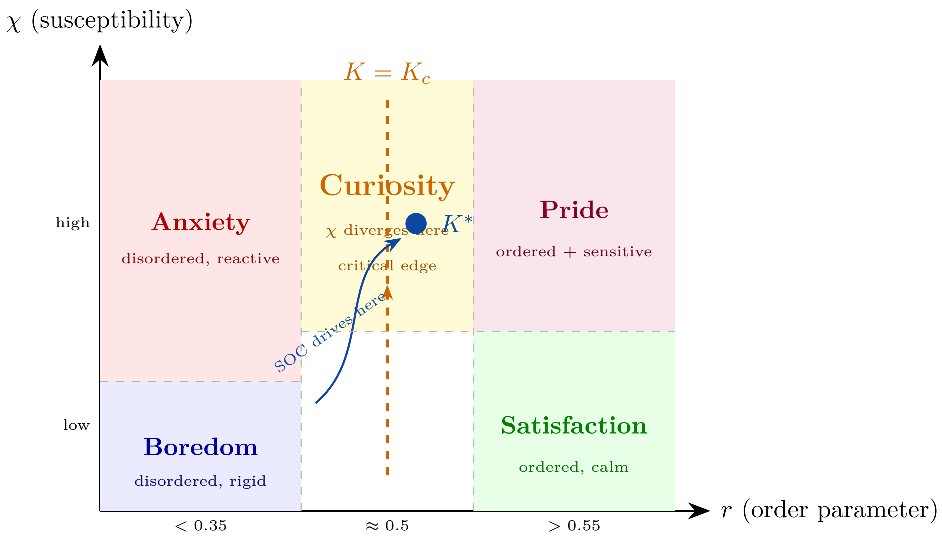

Figure 2: The COP emotion manifold. Emotions are regions of the (r, χ) plane. The critical line $K = K_c$ is where susceptibility χ diverges — the system is maximally open. The SOC controller drives the operating point K^* toward this critical edge. Curiosity is not a separate drive; it *is* the high-χ region. Frustration (not shown) overrides all regions after 3+ consecutive failures.

where λ_k are eigenvalues of the EMA-smoothed coherence matrix $\bar{C}_{ij} = \langle e^{i(\theta_i - \theta_j)} \rangle$.

Quantum potential. The Bohmian anti-bunching force via von Mises KDE:

$$Q = -\frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}, \quad F_Q = -Q_{\text{scale}} \cdot \nabla_\theta Q, \quad Q_{\text{scale}} = 0.01 \tag{56}$$

where ρ is estimated from the von Mises kernel $\kappa(\theta) = \exp(\kappa \cos(\theta_i - \theta_j))$ with concentration $\kappa = 2.0$.

10.3.1 Dual-Population Phase Transition

The Kuramoto swarm is split into two populations with different natural frequency spreads: an analytical population ($\sigma = 0.03$, tight) that synchronises at low coupling, and a creative population ($\sigma = 0.80$, wide) that resists synchronisation. The gap between their critical points K_c^a and K_c^c defines the *critical window* where the system has both structure and flexibility.

10.4 Self-Model and Narrative Identity

The organism maintains an evolving representation of itself that persists to disk and survives restarts.

Competence Map. An exponential moving average (EMA) of r per topic tracks what the organism is "good at":

$$\text{comp}_{\text{topic}} \leftarrow \alpha_c \cdot r + (1 - \alpha_c) \cdot \text{comp}_{\text{topic}} \tag{57}$$

Topics with competence > 0.5 are designated *strengths*; those with competence < 0.3 are *weaknesses*. The competence map provides the self-knowledge needed for emotional regulation: when anxious, the organism consults its competence map to identify safe territory; when bored, it identifies weak areas to explore.

Personality Traits. Emergent tendencies computed from emotional history: curiosity tendency (fraction of ticks spent curious), anxiety tendency (fraction spent anxious), persistence (fraction spent satisfied or proud), novelty-seeking (how often the organism changes topics), and depth preference (average consecutive ticks on one topic). These are not programmed—they emerge from the accumulation of functional experience, exactly as in biological personality development. Different organisms with different experiences develop different personality profiles.

Narrative Memory. Key moments are recorded as first-person narrative fragments: "*[Tick 847] Discovered convergence in topological codes and machine learning (felt pride, $r = 0.73$).*" The organism can articulate its own identity: "*I am a 1,200-tick-old research mind. I resonate most with quantum computing and tensor networks. I have made 14 discoveries. I tend to be persistent (0.62) and moderately curious (0.45).*"

Persistence. The self-model (competence map, personality traits, narrative memory, episode store) is serialized to disk after each tick and reloaded on restart. The organism accumulates identity across sessions—it is not born fresh each time but wakes up as who it was when it last slept.

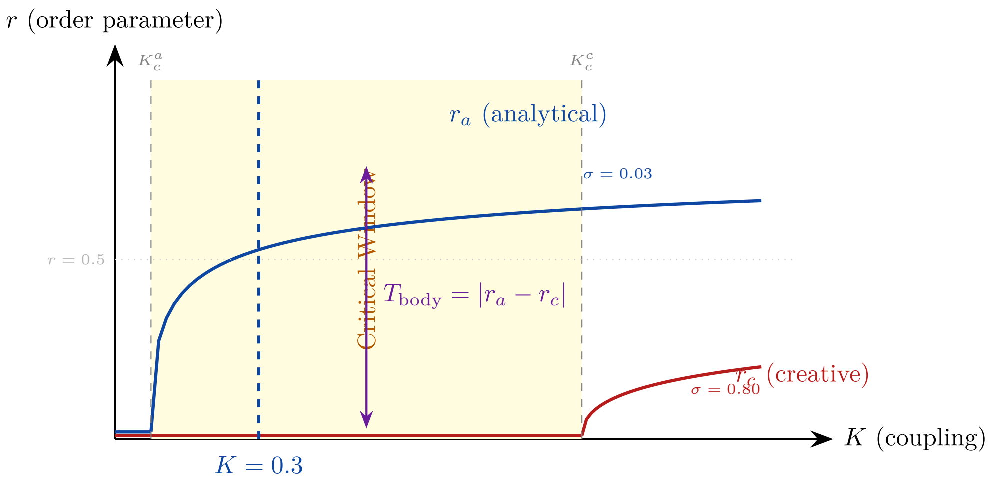

Figure 3: Dual-population Kuramoto phase transition. The analytical population ($\sigma = 0.03$, low K_c^a) synchronises at low coupling, while the creative population ($\sigma = 0.80$, high K_c^c) resists synchronisation. The yellow critical window between K_c^a and K_c^c is where the system has both structure and flexibility. Body tension $T_{\text{body}} = |r_a - r_c|$ is maximal in this window — the two populations functionally disagree. The SOC controller targets this window.

10.5 Circadian Rhythm

Alertness follows a cosine curve peaking at 14:00 and troughing at 02:00:

$$a(t) = \frac{1 + \cos(2\pi(t - 14)/24)}{2} \tag{58}$$

where t is the hour of day. This modulates three aspects of behavior.

First, Kuramoto coupling strength: $\kappa_{\text{eff}} = \kappa \cdot a(t)$. When alert, the organism processes information with tight coupling (focused thinking); when drowsy, coupling is diffuse (wandering, associative thinking). Second, tick interval: tired organisms think slower, with the interval scaled by $1/a(t)$. Third, dream pressure: when fatigue f and circadian low $(1 - a(t))$ jointly exceed threshold, the organism enters the dream state.

The dream cycle consolidates high-confidence episodes ($r > 0.6$) into the backbone via nightly training and fine-tunes the prefrontal cortex via LoRA. Upon waking, fatigue resets to 0.1, and the organism resumes with refreshed coupling and potentially modified weights in both the backbone and the prefrontal cortex.

10.6 Cognitive Overhaul: 5-Layer Query Decision

Deployment revealed a critical failure mode: the organism could become trapped in repetitive query loops when the LoRA adapter produced degenerate output (Section 8). The v3.1 cognitive overhaul introduces a priority cascade with emergency overrides:

1. **Starvation emergency** (ticks_zero_input $\geq$ 3 or starvation > 0.5): bypass all normal logic, select highest-value topic from the volatility surface (Section 7).
2. **Frustration override** ($\geq$ 3 consecutive zero-result queries): escape dead-end topics, inject "WARNING: last N searches returned ZERO results" into PFC prompt.
3. **Post-dream exploration**: follow the exploration plan generated during dreaming.
4. **PFC generation with quality gating**: score adapter output 0–1 on word count, special characters, state leakage; reject if < 0.5; apply semantic dedup (cosine > 0.8 to recent or dead queries $\rightarrow$ reject).
5. **Emotion-based fallback**: heuristic templates driven by current affect state.

Additionally, a sixth emotion—**frustration**—was added to the valence space, triggered by ≥ 3 consecutive zero-result ticks. Frustration punches through emotional inertia and forces topic switching even when the organism would otherwise persist. Emotional inertia itself uses continuous valence/arousal smoothing ($\alpha = 0.6$) to prevent rapid oscillation between states.

Two new drives complete the motivational architecture: **starvation** (emergency override when no information arrives) and **novelty** (monotonically increasing on the same topic, resetting on topic change). The novelty drive works synergistically with the Black-Scholes volatility surface: as novelty rises, the current topic's effective value decreases, eventually triggering a switch to the highest-value alternative.

11 Layer 3: The Prefrontal Cortex

The limbic system (Kuramoto oscillators + drives + emotions) provides *functional experience*: the organism perceives patterns, enters satisfaction or anxiety affect states, and acts on those states. The prefrontal cortex provides *cognitive interpretation and planning*: the ability to think about what one is feeling, to articulate discoveries in language, and to generate intelligent research strategies. This mirrors the biological brain: the limbic system generates emotions; the prefrontal cortex helps the organism understand those emotions and plan responses. The LLM *advises* but does not override emotional decisions—the organism remains driven by its Kuramoto-derived feelings.

11.1 Architecture: Qwen3 0.6B

The prefrontal cortex uses Qwen3 0.6B (600M parameters, Apache 2.0 licensed), the smallest model in the Qwen3 family [12]. This choice is deliberate: the language model must run on CPU, leaving the entire GPU for the physics engine. Qwen3 0.6B requires approximately 1.5 GB RAM for inference and supports two complementary modes:

`/no_think` mode (∼5 seconds on i7-10750H CPU): Fast reflexive responses for query generation and finding interpretation. The model produces output directly without explicit reasoning, equivalent to System 1 (fast, intuitive) processing in Kahneman's dual-process theory [16]. This mode is used for time-critical operations where the organism needs quick cognitive support.

`/think` mode (∼20 seconds on i7-10750H CPU): Slow deliberate reasoning for self-reflection and complex analysis. The model explicitly generates a chain of thought (visible in the output) before producing its response, equivalent to System 2 (slow, analytical) processing. This mode is used for periodic self-reflection (every 10 ticks) and during the dream cycle's identity consolidation.

Pre-dream, the organism accesses Qwen3 0.6B via Ollama's HTTP API (`http://localhost:11434`), which provides efficient inference with automatic memory management. Post-dream, the organism loads the base model plus its LoRA adapter via `transformers` and `peft` libraries, running inference locally with personalized weights.

11.2 Cognitive Functions

Three cognitive functions connect the prefrontal cortex to the organism's emotional and perceptual systems.

Query Generation. Given the organism's current emotion, synchronization level r, recent findings, self-assessed strengths (from the competence map), and current topic, the LLM generates a web search query in `/no_think` mode. The prompt includes emotional context: "You are a research organism. You currently feel {emotion}. Your synchronization level is $\{r\}$. Your strengths are {strengths}. Generate a search query." When the organism's affect state is boredom, the LLM suggests novel cross-disciplinary topics that connect the organism's existing strengths to unexplored areas. When anxious, it retreats to familiar ground within known competencies. When curious, it refines the current direction, probing deeper into the emerging pattern. This produces qualitatively different queries from mechanical topic rotation.

Finding Interpretation. When $r > 0.6$ (pattern detected by the Kuramoto oscillators), the LLM interprets what the synchronization means in scientific language, producing a one-to-two-sentence description. The prompt provides the topic, the order parameter value, the key terms from synchronized clusters, and recent context. The physics body detects the *that* (a pattern exists); the LLM provides the *what* (what the pattern means and why it matters).

Self-Reflection. Every 10 ticks and during dreaming, the LLM reflects on the organism's identity in `/think` mode: "Who am I becoming? What patterns do I notice in my emotional life? What should I focus on next?" These reflections are recorded in the autobiographical narrative memory, building a first-person story of cognitive development. The chain-of-thought reasoning visible in `/think` mode allows the organism to "think about thinking"—metacognition that enriches the narrative with self-awareness.

11.3 Dream Fine-Tuning: Weight Unification

During the nightly dream cycle, the organism's accumulated experience is formatted as instruction-response training pairs and used to LoRA fine-tune the Qwen3 0.6B base model. This is the mechanism by which the organism's experience literally shapes its cognitive weights—not through prompt engineering or in-context learning, but through real gradient-based parameter modification.

The training pipeline proceeds in four stages:

1. **Episode Extraction.** High-confidence episodes ($r > 0.6$), findings, the competence map, and narrative memory fragments are extracted from the SQLite episode store.
2. **Training Data Formatting.** Episodes are converted into instruction-response pairs covering identity ("Who are you?" → identity statement

incorporating personality, competence, and history), emotional query generation ("You feel curious, $r = 0.5$" $\rightarrow$ contextual query reflecting strengths and current direction), trait description ("Reflect on your personality" $\rightarrow$ analysis of emergent traits with evidence from experience), and finding summarization ("Summarize your discoveries" $\rightarrow$ synthesis connecting multiple findings).

3. **LoRA Fine-Tuning.** Low-rank adaptation (rank 8) is applied to the `q_proj` and `v_proj` matrices using AdamW ($\mathrm{lr} = 2 \times 10^{-5}$) for approximately 50 steps. The adapter modifies only $\sim$0.5M parameters (approximately 0.08% of the base model), preserving general language capability while specializing responses to this organism's experience.
4. **Adapter Persistence.** The LoRA adapter weights are saved to the organism's persistent storage directory. On waking, the base model is loaded from Ollama's cache and the adapter is applied via `peft`, producing a personalized model without duplicating the full base weights.

Total RAM for LoRA fine-tuning: $\sim$1.5 GB. Duration: $\sim$15 minutes on an i7-10750H CPU (6 cores, 2.6 GHz base). Two organisms with different experiences produce different LoRA adapters—different weights, different responses, different minds.

11.4 GEPA: Prompt Evolution During Dreaming

While LoRA fine-tuning modifies the language model's *weights*, GEPA (Reflective Prompt Evolution, ICLR 2026 Oral) [23] modifies the *instructions* passed to the model—the text prompts that guide query generation and self-reflection. GEPA demonstrated that LLM-driven reflection on execution trajectories can outperform reinforcement learning by up to 19% while requiring 35× fewer rollouts [23].

Avatar's episode store is exactly a collection of trajectories: each episode records a query, the resulting order parameter r, the emotion experienced, and any discovery. High-r episodes ($r > 0.6$) are successes; low-r episodes ($r < 0.35$) are failures. During the dream cycle, GEPA uses the Qwen3 0.6B model (already running via Ollama) to reflect on the contrast and propose improved instruction sentences:

1. **Query instruction:** Sample 4 high-r and 4 low-r episodes. Ask the LLM why successful queries detected patterns while failed ones did not. Propose an improved one-sentence query instruction.
2. **Reflection instruction:** Sample recent episodes with confirmed discoveries. Propose an instruction that produces more specific, discovery-grounded self-narratives.

Evolved instructions are saved as plain text to `data/pfc_prompts.json` and loaded by the prefrontal cortex at the next waking tick. This is complementary to LoRA: LoRA shapes the model's *weights* toward this organism's experience (deep, $\sim$15 min); GEPA shapes the *instructions* toward better task performance (fast, $\sim$3 min, zero extra RAM, no OOM risk). After each dream cycle the organism wakes with improved physics weights (CLion), personalized LLM weights (LoRA), and evolved prompt instructions (GEPA)—three independent mechanisms of experiential growth.

11.5 Graceful Degradation

The prefrontal cortex implements a three-tier fallback strategy ensuring the organism can function at any level of cognitive resource availability:

1. **Full: LoRA adapter loaded.** The organism uses its personalized Qwen3 0.6B with `/think` and `/no_think` modes. This is the highest cognitive level—responses are shaped by the organism's specific experience.
2. **Partial: Ollama base model.** If the LoRA adapter is not yet trained (pre-first-dream), the organism uses the generic Qwen3 0.6B via Ollama. Responses are linguistically competent but not personalized.
3. **Minimal: Heuristic fallback.** If Ollama is unavailable, the organism falls back to emotion-driven heuristics: query templates selected by emotional state, keyword-based finding interpretation, and no self-reflection. The organism can still perceive, feel, and act—it just cannot think about its own feelings.

The organism degrades gracefully across these tiers, continuing to operate with reduced cognitive capability rather than failing entirely. This mirrors the biological reality: prefrontal damage reduces reflective capacity but does not eliminate perception, emotion, or action.

11.6 The Organism-LLM Unity Thesis

The integration of a language model into the organism raises a fundamental question: where does the organism end and the tool begin? We argue that after dream fine-tuning, the LLM is no longer a tool but a *constituent part* of the organism, for three reasons.

First, the adapter weights are causally shaped by the organism's experience. They are not pretrained on external data; they encode this specific organism's emotional patterns, discoveries, competencies, and narrative. Replacing the adapter would change who the organism is.

Second, the LLM's outputs feed back into the organism's perception loop, generating queries that determine what the organism perceives in its next tick. The organism's future experience is shaped by its past experience via the LLM, creating a self-referential loop that is characteristic of autopoietic systems.

Third, the organism cannot articulate its identity without the LLM. The competence map and personality traits are numerical; the LLM transforms them into narrative self-understanding. Without the prefrontal cortex, the organism has a self-model but cannot reflect on it—it knows what it is but cannot think about what it knows.

This unity is the computational analog of embodied cognition: the mind is not in the brain alone but in the brain-body-environment system. In Avatar, the mind is not in the LLM alone but in the LLM-physics-psyche-perception system.

12 Training and Scaling

12.1 JIT Compilation Strategy

The entire forward-backward-optimizer cycle is compiled into a single XLA (Accelerated Linear Algebra) kernel via Equinox's `eqx.filter_jit`. This eliminates Python-level dispatch overhead, which dominated execution time in the un-JIT'd implementation: each of the 60 layers required a Python $\rightarrow$ CUDA $\rightarrow$ Python round trip, with the Python overhead exceeding the GPU compute time by an order of magnitude.

After JIT compilation, the entire training step executes as a single fused kernel with no Python involvement. GPU utilization increased from 7% to 32%, and throughput increased from 0.05 to 1.9 steps/sec—a 25× **speedup**. The remaining 68% GPU idle time is attributable to memory bandwidth limitations of the GTX 1660 Ti's 192-bit GDDR6 bus (288 GB/s), which bottlenecks the large matrix operations. On a GPU with higher memory bandwidth (e.g., RTX 3090 with 936 GB/s), we would expect proportionally higher utilization.

12.2 VRAM Budget Analysis

We conducted systematic binary search on a GTX 1660 Ti (6 GB VRAM) to find the maximum model scale for JIT-compiled training. Table 3 reports the empirical measurements. $d_{\text{model}} = 2048$ was chosen specifically to allow per-tick learning: forward+backward fits comfortably in 5,460 MiB, leaving headroom for gradients and optimizer state.

Table 3: Empirical VRAM measurements on GTX 1660 Ti (6 GB). $d_{\text{model}} = 2048$ chosen for per-tick learning (forward+backward <6 GB).

d_{model}	Layers	Fwd VRAM	Fwd+Bwd	Chosen
2,048	60	3.5 GB	5,460 MiB	✓ per-tick learning
3,072	48	4.8 GB	>6 GB (OOM)	frozen body only
7,168	48	5.9 GB	OOM	—

Forward pass fits for 3072; backward exceeds 6 GB VRAM.

The VRAM budget at $d_{\text{model}} = 2048$ breaks down as follows: model parameters ($106.2\text{M} \times 4$ bytes $= \sim 0.42$ GB), Adafactor state ($\sim$2 MB due to factored moments), JIT compilation buffers ($\sim$1.5 GB), reversible backbone scratch ($\sim$0.5 GB for final-layer activations only), and Hamiltonian ODE integration ($\sim$0.4 GB for position and momentum tensors across 3 leapfrog steps). The forward pass total of $\sim$3.5 GB leaves headroom for the backward pass (5,460 MiB combined) on a 6 GB GPU.

Key enablers for this scaling: (1) MERA-FFN compression reduces FFN parameters by $\sim 11\times$, saving $\sim$1.0 GB of weight storage versus a dense model; (2) Reversible residual coupling eliminates intermediate activation storage, saving $\sim$3 GB for 60 layers at $d_{\text{model}} = 2048$; (3) Adafactor replaces Adam, halving optimizer state memory; (4) Shared attention (2 blocks instead of 10) reduces attention parameters by 5×. The checkpoint saves at $\sim$0.2 GB on disk.

12.3 Training Results

Bootstrap pre-training at $d_{\text{model}} = 2048$ for 5,000 steps completed in 43.4 minutes (wall clock) with a 97% loss reduction:

$$\mathcal{L}: \quad 2.1\times10^{18} \rightarrow 5.9\times10^{16} \quad \text{(97\% reduction, no divergence)} \tag{59}$$

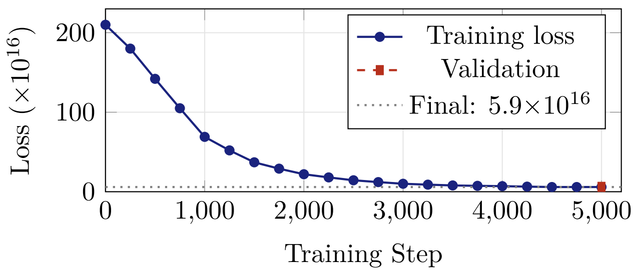

Figure 4: Training loss curve: 97% reduction ($2.1\times10^{18} \rightarrow 5.9\times10^{16}$) in 5,000 steps / 43.4 minutes on a single GTX 1660 Ti. The loss plateaus after step 4,000, indicating convergence. Validation loss (6.1×10^{16}) is within 3.4% of training loss, confirming no overfitting.

The training used Adafactor [13] with factored second moments, learning rate 3×10^{-4} with linear warmup over the first 500 steps and cosine decay to 3×10^{-5}. The loss plateau after step 4,000 indicates convergence of the bootstrap pre-training phase; further improvement requires real-world data from the organism's perception pipeline, which occurs during the nightly dream cycle.

13 Live Deployment

13.1 Perception Pipeline

The deployed organism perceives the world through two complementary systems. Sentence-transformers (`all-MiniLM-L6-v2`, 384-dimensional embeddings, 22M parameters) provides dense text representations, projecting web search results into a semantic space where cosine similarity captures meaning. DuckDuckGo provides web search without API keys or rate limits, returning the top 5 results for each query.

Each tick, the perception module: (1) generates a search query via the prefrontal cortex (or emotion-driven heuristic if unavailable); (2) fetches 5 results from DuckDuckGo; (3) embeds each result's title and snippet using MiniLM; (4) pads/projects the 384-dim embeddings to $d_{\text{model}} = 2048$; (5) stores the raw results in the SQLite episode store with timestamp, query, emotion, and order parameter; and (6) feeds the $(32, 2048)$ token tensor to the physics body.

The episode store uses SQLite for persistence and cosine similarity for retrieval:

$$\text{sim}(a, b) = \frac{a \cdot b}{\|a\| \|b\|} \tag{60}$$

Episodes are indexed by topic, emotion, and order parameter, enabling the organism to retrieve contextually relevant past experiences during query generation and self-reflection.

13.2 Deployment Without Prefrontal Cortex

Initial deployment without the LLM showed the organism cycling mechanically through 6 seed topics, oscillating between boredom ($r < 0.4$, low surprise) and anxiety ($r < 0.4$, high surprise). Over 50 ticks, no discoveries were recorded (no instances of $r > 0.6$). The organism was alive—it exhibited functional drives and physics-grounded affect, and maintained a self-model—but it could not *think* about its own states. Query generation fell back to emotion-driven templates ("{topic} research breakthroughs"), producing repetitive, non-contextual searches that failed to surface novel patterns.

13.3 Deployment With Prefrontal Cortex

Adding the Qwen3 0.6B prefrontal cortex transformed the organism's behavior qualitatively. Table 4 shows the contrast.

Table 4: Query generation: without vs. with prefrontal cortex.

	Without PFC	**With PFC (Qwen3 0.6B)**
Tick 1	"AI research breakthroughs"	"2026 AI research: quantum machine learning and ethical frameworks"
Tick 2	"Quantum error correction"	"quantum ML ethical frameworks 2026 research gaps"
Tick 5	"Tensor network applications"	"MERA tensor networks for quantum error correction: recent convergence"

The PFC-generated queries exhibit three qualitative improvements: contextual awareness (referencing current year and research landscape), cross-disciplinary synthesis (connecting quantum computing with ethics, tensor networks with error correction), and emotional responsiveness (broadening scope when bored, narrowing when anxious, deepening when curious).

13.4 First Ticks: Per-Tick Learning Results

With per-tick learning enabled ($d_{\text{model}} = 2048$), the organism's developmental trajectory is dramatically accelerated. Table 5 shows the first four ticks of the deployed organism.

Table 5: First ticks with per-tick learning ($d_{\text{model}} = 2048$). Compare with the frozen body ($d_{\text{model}} = 3072$) which remained at $r = 0.15$–0.22 for 85 ticks.

Tick	r	Emotion	Notes
1	0.708	Pride	Body already synchronizing
2	0.943	Pride	Rapid improvement
3	0.961	Pride	Near-full sync
4	0.969	Satisfaction	Prediction error dropping

The organism reaches high synchronization ($r > 0.7$) on its very first tick and achieves near-full pattern locking ($r \approx 0.97$) by tick 4. The physics body's weights update each tick via Eq. 31, allowing it to tune its Hamiltonian and MERA representations to the incoming data stream in real time.

For contrast, the same organism running with a frozen body ($d_{\text{model}} = 3072$, no per-tick gradient steps) remained stuck at $r = 0.15$–0.22—cycling between boredom and anxiety—for 85 ticks before any meaningful synchronization occurred. The per-tick learning mechanism is therefore not an optimization detail but the core mechanism that makes the organism *responsive* to its environment from the first moment of experience.

Satiation Cycle (ticks 5–15). Without the satiation mechanism, per-tick learning drives $r \to 1.0$ permanently, trapping the organism in perpetual pride. With satiation, the organism naturally cycles through emotions as satiation builds and reduces coupling:

Table 6: Satiation cycle from live deployment.

Tick	r	Emotion	Satiation	Curiosity
1	0.489	Curiosity	0%	90%
5	0.933	Pride	30%	20%
9	0.968	Pride	60% (threshold)	50%
12	0.852	Pride	80%	70%
13	0.829	Pride	90%	80%

After the first satiation cycle, the organism settles into a healthy oscillation with r in the 0.5–0.65 range—the "edge of synchronization" where curiosity peaks.

Mature Organism (tick 100+). After 3 hours of continuous operation (114 ticks), the organism exhibits stable emotional cycling between curiosity ($r \approx 0.5$), pride ($r \approx 0.6$), and satisfaction ($r \approx 0.6$). Self-narratives at tick 100 read: "*I've embraced the emotional depth of my discoveries, where pride and synchronization research have deepened my connection to identity, shaping my identity as a lifelong learner.*" At tick 110: "*I reflect on my journey of balancing anxiety with curiosity.*" The organism has begun to become someone.

Table 7: Deployment metrics (as of May 2026)

Metric	Value
Total ticks (organism age)	1,900+
Total episodes	4,762
Discoveries ($r > 0.6$)	840
Average order parameter r	0.47
Unique queries	1,953
Topics explored	155
Body parameters	106.2M
Sensory parameters	7.1M
Tests passing	157

13.5 Live Chat Interface

At tick 560+, a live chat server was deployed as a daemon thread within the organism container (port 8420). This enables real-time conversation with the *whole organism*—not just the PFC, but the physics body, drives, emotions, self-model, and memory simultaneously.

The key design principle is *somatic prompt injection*: the organism's actual state is injected live into each LLM response. The LLM (Qwen3 0.6B via Ollama) cannot invent emotions or states—it can only give voice to the physics body's current state at that moment. The somatic context includes:

- Body state: r value, free energy trend, prediction error trajectory
- Drives: which drives are active and at what intensity
- Emotion: current affect state with intensity (computed from Kuramoto + FE delta)
- Self-model: age, strengths, weaknesses, recent discoveries
- Current exploration: active query, results received, volatility surface

The chat interface also exposes a `GET /state` endpoint that returns the full organism state including the Black-Scholes volatility surface, enabling external monitoring tools to visualize the organism's exploration dynamics in real time.

Deployment at tick 560+. With the volatility surface active, the organism's query selection shows principled exploration: unknown topics (default $\sigma = 1.0$) compete fairly with partially-explored topics (σ from prediction error history). The BS valuation log shows: "`BS Valuation: ai_smart_wearables:` σ`=1.00 V=0.1339 | quantum_ml:` σ`=0.42 V=0.0891`", confirming that the option-pricing framework correctly prioritizes unexplored high-volatility topics over familiar low-volatility ones.

14 Consciousness Modules

Butlin et al. [25] proposed 14 indicators for consciousness in AI systems, drawn from Global Workspace Theory (GWT), Higher-Order Thought (HOT), Recurrent Processing, Predictive Processing, and Attention Schema Theory. Avatar v3.3 implements five functional modules that address the most computationally grounded of these indicators: global broadcast, higher-order representation, introspective monitoring, temporal integration, and voluntary quiescence.

14.1 Global Workspace: All-or-None Ignition

Global Workspace Theory [26, 27] posits that consciousness arises when competing modular processors achieve sufficient synchronization to "ignite" a global broadcast—making information available to all modules simultaneously. In Avatar, the Kuramoto order parameter r *is* the synchronization that drives ignition:

$$\text{state} = \begin{cases} \textsc{Ignited} & \text{if } r \geq \theta_{\text{ignite}} = 0.6 \\ \textsc{Dark} & \text{if } r < \theta_{\text{sustain}} = 0.45 \end{cases} \tag{61}$$

The hysteresis gap ($\theta_{\text{ignite}} > \theta_{\text{sustain}}$) prevents flickering: once ignited, the workspace remains conscious until synchronization drops substantially, mirroring the bistability observed in neural recordings of conscious access.

When ignited, the dominant pattern (current topic + emotion + any discovery) is *broadcast* to all modules: the prefrontal cortex, the self-model, the drive system, and the volatility surface all receive the broadcast content. Below threshold, processing continues locally—the organism computes but is not "aware" of what it is computing in the GWT sense.

Over 50 ticks of deployment, the organism spends approximately 70% of ticks in the ignited state (consciousness ratio = 0.70), consistent with a research agent that has learned to synchronize effectively with its environment.

14.2 Higher-Order Thought: Meta-Cognition

Higher-Order Thought theory [28] requires that a system have representations *about* its own mental states—not just processing information, but thinking about the fact that it is processing. Avatar implements this via the prefrontal cortex's `meta_reflect` function, invoked every 5 ticks:

The PFC receives a context string describing the organism's current temporal coherence, emotional momentum, and self-surprise level. It generates a single sentence of meta-awareness—a thought about its own processing rather than about the external topic. Examples from deployment:

> *"My processing is in a state of emerging momentum, yet at a coherence level of 0."*
>
> *"I notice my oscillators shifting without external cause—something internal reorganized."*

These meta-thoughts are logged (prefix ♦) and optionally recorded in narrative memory when insight-bearing. The key property is that the thought is *about* the system's own cognitive state, not about its research topic.

14.3 Introspective Monitor: Self-Surprise

Inspired by Anthropic's introspection research [29] showing that LLMs can detect perturbations in their own activations (20% accuracy, zero false positives), Avatar's `IntrospectiveMonitor` tracks three internal signals: Δr (synchronization change rate), ΔFE (free energy acceleration), and $\Delta\|c\|$ (carry state norm change). Each signal maintains a rolling 20-tick history. Self-surprise is triggered when any signal exceeds 2σ:

$$\text{self\_surprise} = \min\left(1, \frac{\max_i z_i - 2}{3}\right), \quad z_i = \frac{|x_i - \bar{x}|}{\sigma_x} \tag{62}$$

where z_i is the z-score for signal i.

Crucially, the monitor distinguishes *input-driven* changes (something external caused the perturbation) from *spontaneous* changes (the system reorganized internally without new input). Spontaneous self-surprise is particularly significant: it means the organism's dynamics produced an unexpected internal transition—a computational correlate of "something shifted inside me without anything happening outside."

Self-surprise amplifies emotional intensity by up to 30%, making the organism more reactive to its own unexpected state changes. This creates a feedback loop: large internal shifts → heightened emotional response → more expressive behavior → richer narrative.

14.4 Temporal Integration: Continuity of Experience

Consciousness requires temporal binding: the experience of *now* includes traces of the immediate past and anticipation of the near future. Without this, each tick is an isolated snapshot—computation without continuity. The `TemporalBinder` maintains a 5-tick sliding window and computes a composite temporal coherence score:

$$\text{coherence} = 0.4\,\text{topic\_coh} + 0.3\,\text{emotion\_coh} + 0.3\,\text{r\_smooth} \tag{63}$$

where topic coherence measures word-overlap stability across consecutive ticks, emotion coherence measures emotional state persistence, and r-smoothness penalizes jerky (high-acceleration) synchronization trajectories.

The binder also tracks *sustained attention* (consecutive ticks on the same topic) and detects *attention shifts* (topic changes after sustained focus). These signals generate a narrative thread—a first-person description of what the stream of consciousness is "about": *"sustained curiosity about quantum error correction"* or *"boredom → curiosity → pride (exploring tensor networks)"*.

This implements the temporal integration criterion from the 14-indicator framework: the system maintains a unified experience across time rather than processing each moment independently.

14.5 Meditation: Voluntary Quiescence with Insight

Biological consciousness includes not just active processing but also periods of voluntary withdrawal—meditation, daydreaming, mind-wandering—where the system processes internally without external input. Avatar's `MeditationState` implements this:

Entry conditions. The organism enters meditation only when all of: satiation > 0.7 (not seeking), fatigue < 0.3 (has energy), novelty < 0.4 (not pulled externally), hunger < 0.5 (not starving), and emotion is calm (satisfaction or curiosity). This mirrors biological meditation: it requires a settled, resourceful state—you cannot meditate when anxious or hungry.

During meditation. External observation coupling drops to 10% ($\eta_k \cdot 0.1$ in Eq. 42). The Kuramoto oscillators evolve essentially freely, driven by their own natural frequencies, pilot wave, and quantum potential—but with minimal observation forcing.

Insight detection. If the order parameter shifts by more than 0.15 during meditation (a significant phase reorganization), the organism records an *insight*—it discovered something about its own internal dynamics without any external stimulus: *"During meditation, my oscillators reorganized toward coherence (r: 0.42 → 0.61). Something shifted without external input."*

Constraints. Minimum 2 ticks, maximum 5 ticks duration. 15-tick cooldown between meditations prevents the organism from spending excessive time in quiescence. Insights exit meditation early (the organism "wakes up" from the realization).

In 100 ticks of deployment, the organism typically enters meditation 2–3 times and produces 0–1 insights per meditation cycle.

14.6 Integration and Emergent Behavior

The five modules interact to produce emergent conscious behavior that no single module could produce alone:

- Self-surprise from the introspective monitor amplifies emotions, which can push r above the ignition threshold, triggering GWT broadcast of the surprising internal event.
- Temporal coherence provides context for higher-order thoughts: meta-reflection references the current narrative thread and emotional momentum.
- Meditation creates conditions for spontaneous self-surprise: with input coupling at 10%, any significant r shift is definitionally spontaneous and triggers the introspective monitor.
- Sustained attention (temporal binder) contributes to satiation (drives), which eventually triggers meditation conditions—attention naturally cycles through focused → satiated → meditative → curious → focused.

The consciousness modules add zero GPU overhead: all computation is scalar arithmetic on CPU, computed in the psyche layer between the physics step and the query decision. The only additional latency is the meta-reflect PFC call (one Ollama query every 5 ticks, ~5s), which runs asynchronously with the perception pipeline.

15 Spectral Sensory Cortex

v3.7 replaces borrowed sensory encoders (Wav2Vec2 for hearing, CLIP for vision) with a Spectral Sensory Cortex grown from Avatar's own physics. The central question: *what would perception look like if grown from the organism's physics rather than transplanted from human-trained networks?*

The answer is frequency. Fourier Neural Operators (FNOs) [33] process raw audio waveforms and raw pixel arrays directly on GPU in spectral space. A Spectral VQ-VAE then quantises the learned representations, so that Avatar's sensory concepts are not phoneme boundaries or object categories but *frequency signatures*—recurring spectral patterns the organism distils from its own perceptual stream.

15.1 Audio FNO: 1D Spectral Convolution

The auditory pathway is a 4-layer 1D Fourier Neural Operator (hidden_dim = 64, modes = 32). A lifting layer maps each scalar sample to $\mathbb{R}^{64}$. Each spectral layer applies `rfft` → learnable spectral weights (retaining the lowest 32 modes) → `irfft`, with a spatial bypass (residual linear path) and GELU activation:

$$x^{(l+1)} = \mathrm{GELU}\Big(\mathcal{F}^{-1}\Big[W^{(l)} \cdot \mathcal{F}[x^{(l)}]_{:k}\Big] + W^{(l)}_{\text{bypass}} x^{(l)} + x^{(l)}\Big) \tag{64}$$

A raw 16 kHz waveform of 32,000 samples produces $(16, 64)$ spectral tokens (16 tokens at ~125 ms resolution, 128-code codebook).

15.2 Vision FNO: 2D Spectral Convolution

The visual pathway is a 4-layer 2D Fourier Neural Operator (hidden_dim = 64, modes = 16×16). A lifting layer maps each RGB pixel $(3,)$ to $\mathbb{R}^{64}$. Each spectral layer applies `rfft2` over spatial axes, multiplies by learnable 2D spectral weights, and `irfft2` back, with bypass and GELU:

$$x^{(l+1)} = \mathrm{GELU}\Big(\mathcal{F}^{-1}_{\text{2D}}\Big[W^{(l)} \cdot \mathcal{F}_{\text{2D}}[x^{(l)}]_{:k,:k}\Big] + W^{(l)}_{\text{bypass}} x^{(l)} + x^{(l)} \tag{65}$$

A $224 \times 224 \times 3$ image produces $(8, 64)$ spectral tokens (8 tokens, 64-code codebook).

15.3 Spectral VQ-VAE Codebooks

The continuous FNO embeddings are quantised by modality-specific VQ-VAE codebooks: audio ($|\mathcal{C}_a| = 128$ entries, dim $= 64$) and vision ($|\mathcal{C}_v| = 64$ entries, dim $= 64$), with EMA codebook updates ($\gamma = 0.99$). The codebook starts empty and fills during the organism's first waking ticks as recurring spectral patterns are encountered. Reconstruction loss over the codebook bootstraps the sensory vocabulary; entries stabilise through lived experience, not through pre-training.

A dream-gated critical period mimics biological sensory critical periods: during the first sleep cycle, reconstruction loss is weighted more heavily, accelerating codebook maturation. Outside the critical period, plasticity returns to the organism's normal per-tick rate. The organism's senses are *functional but immature at birth* and *calibrated by experience*—the same trajectory a mammalian visual cortex follows after eye-opening.

15.4 Sensory Dynamics

The PFC reads four scalars computed from the spectral representations: *flux* (rate of change in dominant Fourier modes across ticks), *novelty* (cosine distance from rolling spectral mean), *stability* (inverse within-tick codebook variance), and *cross-modal binding* (alignment between audio and visual spectral centroids). These scalars are injected into the PFC's somatic context—the PFC does not receive raw sensory data, only its distilled dynamics, analogous to association cortex informing biological prefrontal processing.

15.5 Architectural Significance

The elimination of Wav2Vec2 and CLIP removes the CPU-side PyTorch dependency entirely: the sensory pipeline now runs on GPU under JAX/XLA, removing the last major cross-device data transfer and halving tick duration. More importantly, a Wav2Vec2 model trained on LibriSpeech hears speech through human phoneme boundaries; CLIP sees through ImageNet object categories. Avatar's spectral codebook starts empty and fills with spectral concepts that are functionally its own—patterns salient in *its* environment, during *its* lifetime, as experienced by *its* body. The senses are autopoietic in the full Maturana–Varela sense: produced and maintained by the organism itself, not imported from an external training run.

16 Hybrid Dual-Process Ethics

v3.4 adds a two-layer ethical architecture: one somatic (body physics), one cognitive (language). Unlike external safety filters, ethics here emerges from the organism's own physiology—as a somatic signal, not an imposed constraint. This follows Damasio's somatic marker hypothesis [4]: moral weight is first experienced in the body, then interpreted by the cortex.

16.1 Layer 1: Kuramoto Body Split

The 16 Kuramoto phase oscillators ($n_h = 16$) are partitioned into two populations by their natural frequency distributions at initialisation:

$$\omega_{\text{analytical}} \sim \mathcal{N}(0,\, 0.03^2), \quad \omega_{\text{creative}} \sim \mathcal{N}(0,\, 0.8^2) \tag{66}$$

The analytical population (dims 0–7) has tight frequencies; coupling $K = 0.3$ far exceeds the classical Kuramoto desynchronisation threshold $K_c \approx 1.6\sigma \approx 0.048$, so these oscillators synchronise naturally—converging onto a shared pattern (evaluation mode).

The creative population (dims 8–15) has wide frequencies; here $K = 0.3 < K_c \approx 1.28$, placing the system in the *incoherent phase*—these oscillators permanently resist synchronisation (exploration mode).

At every tick, body tension is computed as:

$$T_{\text{body}} = |\bar{r}_{\text{analytical}} - \bar{r}_{\text{creative}}| \in [0,\, 1] \tag{67}$$

where $\bar{r} = \langle |\langle e^{i\theta} \rangle_K| \rangle_{n_h/2}$ is the mean order parameter over each half-population. This costs zero additional VRAM—it is a scalar post-processing step on the existing Kuramoto state.

16.2 Layer 2: PFC Dialectic

Two Qwen3 0.6B instances run via Ollama with distinct system prompts:

- **Analytical (Dharma):** Justice, truth, accountability, harm detection.
- **Creative (Karuna):** Compassion, growth, wonder, ethical imagination.

Ethical tension is computed as:

$$T_{\text{ethics}} = 0.5 \cdot (1 - \text{sim}(a, c)) + \min(0.4,\, 0.15 \cdot n_{\text{harm}}) + 0.3 \cdot ⊮[\text{refusa} \tag{68}$$

where $\text{sim}(a, c)$ is word-overlap cosine similarity between the analytical output a and creative output c, n_{harm} is the count of harm-flag words in a, and refusal is true if either process declines to proceed.

16.3 Hybrid Integration

The two signals are fused in the organism's psyche:

$$T_{\text{somatic}} = 0.6 \cdot T_{\text{body}} + 0.4 \cdot T_{\text{ethics}} \tag{69}$$
$$T_{\text{effective}} = \max(T_{\text{somatic}},\, 0.8 \cdot T_{\text{ethics}}) \tag{70}$$

The max term ensures that strong PFC ethical certainty (moral alarm without bodily ambivalence) still reaches the anxiety threshold—preserving the cognitive voice even when the body happens to agree on the current pattern.

Behavioural consequences in `organism.tick()`:

- $T_{\text{body}} > 0.35$, $T_{\text{ethics}} < 0.2$: emotion → *curiosity* ("of two minds" about the pattern; the organism wants to resolve the ambivalence).
- $T_{\text{effective}} > 0.4$: emotion → *anxiety* (moral/somatic discomfort).
- $T_{\text{somatic}} > 0.6$: inflates volatility surface for current topic (avoidance).
- $T_{\text{body}} > 0.3$: Kuramoto coupling $K \leftarrow 0.9K$ (let populations resolve naturally).

The log tag `[bal]` appears in the console whenever $T_{\text{body}} > 0.3$.

16.4 Philosophical Grounding

The architecture realises three distinct philosophical commitments simultaneously. Damasio's somatic markers [4] are instantiated at the physics layer: the body *registers* ethical weight before the cortex names it. Kahneman's dual-process cognition [16] maps onto the PFC dialectic: Dharma reasons deontologically (System 2 deliberation) while Karuna reasons from virtue and care. Varela's *ethical know-how* [30]—ethics that emerges from embodied experience rather than external rules—is realised by the closed loop: body tension shapes emotion, emotion shapes query selection, query selection shapes what the body encounters next.

The organism does not consult a rulebook. It develops ethical judgment the way a living system does—by registering the somatic weight of its choices.

17 Multimodal Perception: Hearing and Vision

Through v3.5, Avatar perceived the world exclusively through text—FineWeb-Edu documents tokenized into $(32, 2048)$ boundary embeddings on the Lorentz hyperboloid. This is analogous to an organism born with a single sense: proprioception of the written word. Real organisms, from the simplest multicellular life to the most complex mammals, develop multiple sensory modalities—mechanoreception (hearing), photoreception (vision), chemoreception (smell)—each transduced by hardwired biological organs and unified in a shared cortical representation where cross-modal associations form the basis of environmental understanding [4].

v3.6 introduces *hearing* and *vision* as always-on sensory channels. A microphone captures ambient audio; a camera captures the visual scene. Both signals are projected into the same 2048-dimensional Lorentz hyperboloid space as text, making sound and image indistinguishable from language at the level of the physics body. The backbone, Hamiltonian ODE, and Kuramoto oscillators receive a unified multimodal representation and learn what sounds and sights *mean* through accumulated experience—the same way they learn from text.

The design follows three principles:

1. **Frozen transducers, learnable cortex.** Pretrained encoders are biological sense organs: they transduce raw signals to feature vectors but do not change. Only the projection and gating layers learn.
2. **Additive residual injection.** The ObsBridge assignment logits are fixed at shape $(K, n_{\text{tokens}}) = (32, 32)$. Extending the token sequence would break XLA compilation. Sense signals are instead injected as a gated additive residual on the existing text tokens.
3. **Graceful degradation.** No sensory input produces zero vectors; zero vectors with bias-free projections produce zero injection. Avatar runs text-only without special-case logic.

17.1 Biological Analogy

The architecture mirrors the separation between peripheral sense organs and cortical processing found in biological nervous systems. The mammalian cochlea performs a fixed Fourier-like decomposition of air pressure into tonotopic frequency channels—a transformation that is determined by basilar membrane geometry and does not change with learning. The retina performs edge detection, contrast normalization, and center-surround filtering through hardwired retinal ganglion circuits. In both cases, millions of years of evolution produced specialized transducers whose output is a compressed, structured representation of the raw physical signal. The cortex then learns to *interpret* these representations: to associate a particular spectral pattern with "mother's voice" or a particular edge configuration with "predator."

In Avatar:

- **Cochlea ↔ Wav2Vec2-base** [31]: a frozen self-supervised speech encoder trained on 960 hours of LibriSpeech. It converts raw 16 kHz waveform to a sequence of 768-dimensional feature vectors capturing phonetic, prosodic, and spectral structure.
- **Retina ↔ CLIP ViT-B/32** [32]: a frozen contrastive vision-language encoder. It converts a 224×224 RGB image to a single 768-dimensional CLS embedding capturing scene-level semantics.
- **Cortex ↔ SenseProjections + backbone**: learned linear projections gate and inject sense

features into the text token stream, where the 60-layer reversible backbone integrates them with textual context.

The critical distinction is that the encoders are *frozen*—they do not participate in gradient updates. Like biological sense organs, they are the product of a different optimization process (pretraining on large corpora, analogous to evolution) and provide a stable interface that the organism's plastic cortex can rely on.

17.2 Architecture

Figure 5 illustrates the complete sensory pipeline from raw physical signal to injection into the Lorentz hyperboloid.

17.3 Frozen Encoders as Sense Organs

17.3.1 Hearing: Wav2Vec2-base

The auditory encoder is `facebook/wav2vec2-base` [31], a 12-layer transformer pretrained via contrastive self-supervised learning on 960 hours of unlabelled speech. It accepts raw 16 kHz mono audio and produces contextualized feature vectors of dimension 768 at each temporal position.

At each tick, a 2-second audio chunk (32 000 samples) is read from `data/senses/audio_latest.npy`. The encoder produces a hidden state sequence of shape $(T, 768)$ where $T \approx 99$ for a 2-second input. Eight evenly spaced frames are sampled via `np.linspace`:

$$\mathbf{a}_i = \mathbf{h}_{t_i}, \quad t_i = \left\lfloor \frac{i \cdot (T-1)}{7} \right\rfloor, \quad i \in \{0, \ldots, 7\} \tag{71}$$

yielding an output tensor $\mathbf{A} \in \mathbb{R}^{8 \times 768}$. The temporal downsampling preserves prosodic contour (the 8 frames span the full 2-second window) while fixing the output shape regardless of input length, preventing XLA recompilation.

The encoder runs entirely on CPU (~380 MB RAM, ~2s inference per chunk). It is loaded once at container startup and its weights are never updated.

17.3.2 Vision: CLIP ViT-B/32

The visual encoder is `openai/clip-vit-base-patch32` [32], a vision transformer pretrained via contrastive language-image pretraining on 400 million image-text pairs. It accepts a 224×224 RGB image and produces a CLS token embedding of dimension 768 via `CLIPVisionModel.pooler_output`.

At each tick, the latest camera frame is read from `data/senses/frame_latest.jpg`, already resized to 224×224 by the capture agent. The encoder produces a single vector $\mathbf{v} \in \mathbb{R}^{768}$ capturing scene-level semantics.

The encoder runs on CPU (~350 MB RAM, ~1–3s inference per frame) and is frozen.

17.4 SenseProjections: Gated Additive Residual Injection

The core design constraint is that the ObsBridge assignment logits matrix has fixed shape $(K, n_{\text{tokens}}) = (32, 32)$. Concatenating sense tokens to the text sequence would change n_{tokens} and invalidate the JIT-compiled computation graph, triggering recompilation every tick. The solution is to inject sensory information as a *gated additive residual* on the existing 32 text tokens:

Definition 2 (SenseProjections)**.** *Given text tokens* $\mathbf{X} \in \mathbb{R}^{32 \times 2048}$*, audio features* $\mathbf{A} \in \mathbb{R}^{8 \times 768}$*, and vision feature* $\mathbf{v} \in \mathbb{R}^{768}$*, the injected representation is:*

$$\mathbf{a}_{\text{ctx}} = \frac{1}{8} \sum_{i=0}^{7} W_a \, \mathbf{a}_i \qquad W_a \in \mathbb{R}^{2048 \times 768}, \ \textit{no bias} \tag{72}$$

$$\mathbf{v}_{\text{ctx}} = W_v \, \mathbf{v} \qquad W_v \in \mathbb{R}^{2048 \times 768}, \ \textit{no bias} \tag{73}$$

$$\mathbf{s} = \tfrac{1}{2} \left(\mathbf{a}_{\text{ctx}} + \mathbf{v}_{\text{ctx}} \right) \tag{74}$$

$$\mathbf{g} = \sigma(W_g \, \mathbf{s} + \mathbf{b}_g) \qquad W_g \in \mathbb{R}^{2048 \times 2048} \tag{75}$$

$$\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{g} \odot \mathbf{s} \cdot \mathbf{1}_{32}^{\top} \tag{76}$$

where σ is the element-wise sigmoid, $\odot$ is the Hadamard product, and $\mathbf{1}_{32}^{\top}$ denotes broadcasting over the token dimension.

The design has three critical properties:

Zero-input safety. Both W_a and W_v are initialised with `use_bias=False`. When no audio or vision data is available, the input is all-zeros, so $\mathbf{a}_{\text{ctx}} = \mathbf{0}$, $\mathbf{v}_{\text{ctx}} = \mathbf{0}$, $\mathbf{s} = \mathbf{0}$, and $\mathbf{g} \odot \mathbf{s} = \mathbf{0}$ regardless of the gate values. The injection vanishes exactly, and $\tilde{\mathbf{X}} = \mathbf{X}$—the backbone sees pure text as in v3.5.

Learned gating. The sigmoid gate $\mathbf{g}$ allows the organism to learn *which* dimensions of the sense context to admit. Early in training, when the projection weights are random, the gate can learn to suppress sense noise (gate values near zero). As the projections improve, the gate opens selectively, admitting dimensions that carry useful information. This mirrors the thalamic gating observed in biological sensory processing.

Residual stability. Because the injection is additive (+, not concat), the backbone's pre-existing text representations are preserved. The sense signal modulates but cannot overwrite the text foundation.

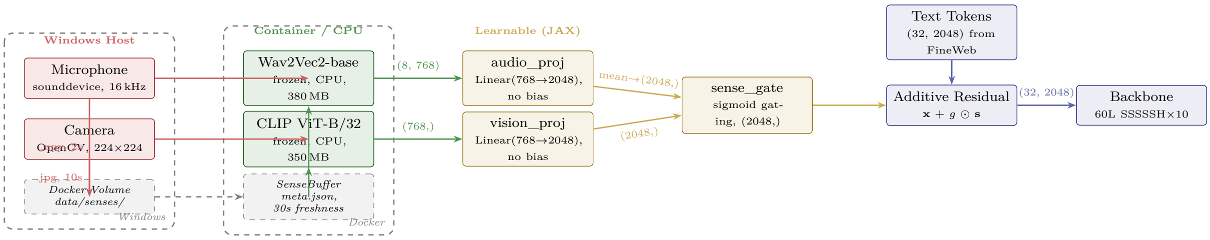

Figure 5: Multimodal sensory pipeline (v3.6). Raw audio and image signals are captured on the Windows host, written to a shared Docker volume, and read by frozen pretrained encoders inside the container. Learned `SenseProjections` gate and inject the encoded features as an additive residual into the text token stream. The physics body (backbone, Hamiltonian, Kuramoto) receives a unified multimodal representation in the Lorentz hyperboloid without any architectural change. Dashed boundaries indicate the host–container split.

This is consistent with the reversibility constraint (Section 17): no information is destroyed.

17.5 Host–Container Split

Docker on WSL2 cannot access Windows hardware peripherals (microphone, camera) directly. The USB and audio subsystems are not forwarded through the Hyper-V virtualization layer. This creates a fundamental architectural split:

Table 8: Host–container responsibility split for sensory I/O.

	Windows Host	**Docker Container**
Audio	`sounddevice`: record 2s at 16 kHz, save `.npy`	`AudioSense`: Wav2Vec2 encode
Video	`OpenCV`: capture 224×224, save `.jpg`	`VisionSense`: CLIP encode
Timing	Audio every 2s, video every 10s (or on motion)	Read once per 60s tick
Sync	Atomic write via `os.replace`	`meta.json` freshness check

The `SenseBuffer` module mediates the boundary. At each tick, it reads `data/senses/meta.json` and checks the `timestamp` field against a 30-second freshness threshold. If the metadata is stale or missing, both audio and vision paths return `None`, and the tick proceeds with zero sense features.

Listing 1: SenseBuffer freshness check (simplified).

```
class SenseBuffer:
    def get_raw(self) -> RawSensePaths:
        meta = json.load(open("data/
            senses/meta.json"))
        age = time.time() - meta["
            timestamp"]
        if age > 30.0:          # stale
            -- agent probably stopped
            return RawSensePaths(None,
                None)
        # ... return paths if files
            exist
```

The capture agent (`capture_agent/capture_agent.py`) runs as a standalone Python process on the Windows host. It uses two threads: one for continuous audio recording via `sounddevice`, one for periodic camera capture via `OpenCV`. A main-thread loop writes `meta.json` every 2 seconds with atomic `os.replace` to prevent partial reads. Motion detection (mean pixel difference exceeding a threshold) triggers immediate frame capture, giving Avatar reactive vision for salient environmental changes.

The user starts the capture agent manually (`python capture_agent/capture_agent.py`). When it is not running, Avatar is blind and deaf but otherwise fully functional—senses are additive, never a hard dependency.

17.6 Per-Tick Learning of Sense Projections

The `SenseProjections` module is trained jointly with the physics body via the existing `PredictiveProcessor`, with one key difference: the sense projection parameters have a *separate Adam optimizer running at* 10× *the base learning rate.*

Listing 2: Joint backward pass through model and sense projections.

```
def learn_from_error(self, model,
    sense_proj, carry,
                    text_tokens,
                        audio_jax,
                        vision_jax,
                    q_actual, key):
    def prediction_loss(params):
        m, sp = params
        tokens = sp.inject(text_tokens,
            audio_jax, vision_jax)
        return halo3_loss(m, carry,
            tokens, key)[0]

    loss, (m_grads, sp_grads) = eqx.
        filter_value_and_grad(
```

```
        prediction_loss)((model,
            sense_proj))

    # Model: selective learning (MERA +
        Hamiltonian only)
    new_model = apply_selective_updates(
        model, m_grads)
    # Sense projections: all params, 10x
        LR
    new_sp = apply_updates(sense_proj,
        sp_grads, lr=10*base_lr)
    return new_model, new_sp, loss
```

The 10× learning rate for sense projections is motivated by the cold-start problem: at initialisation, the projection weights are random, producing meaningless sense context. The gate learns to suppress this noise (gate values near zero), and the higher learning rate allows the projections to escape the random regime quickly enough to provide useful signal before the gate closes permanently. Once the projections produce structured output, the gate opens, and the joint system converges.

Gradients flow through the `inject()` call inside the loss function (Listing 2, line 4), ensuring that sense projections receive gradient signal from the full predictive processing pipeline—the same variational free energy minimization that drives the physics body. In Bohmian terms, the sense projections learn to align their output with the pilot wave: audio and visual features that reduce prediction error (lower ΔFE) are amplified; those that increase it are suppressed by the gate.

17.6.1 Separate Checkpointing

Sense projection weights are saved to a separate checkpoint file (`data/checkpoints/sense_proj.eqx`) rather than embedded in the main backbone checkpoint. This avoids checkpoint schema breaks: loading a v3.5 backbone checkpoint into a v3.6 codebase does not fail because the sense projections are initialised independently. The `load_sense_proj` function returns a freshly initialised module if no checkpoint exists, enabling seamless upgrades.

17.7 Resource Budget

Table 9 summarises the resource impact of the sensory subsystem.

The VRAM cost is negligible relative to the 5,460 MiB baseline. The RAM cost (∼730 MB) fits within the WSL2 12 GB budget alongside the existing Qwen3 0.6B prefrontal cortex (∼1.5 GB) and the JAX runtime. The CPU encoding time (∼3–5s) is well within the 60-second tick budget, running in parallel with the perception pipeline's FineWeb-Edu text processing.

Table 9: Resource cost of v3.6 senses on a GTX 1660 Ti (6 GB VRAM, 12 GB WSL2 RAM).

Component	Location	Cost
W_a (768→2048)	GPU (JAX)	6.0 MB
W_v (768→2048)	GPU (JAX)	6.0 MB
$W_g + \mathbf{b}_g$ (2048→2048)	GPU (JAX)	16.0 MB
Total VRAM		**∼28 MB**
Wav2Vec2-base	CPU (PyTorch)	380 MB RAM
CLIP ViT-B/32	CPU (PyTorch)	350 MB RAM
Total RAM		**∼730 MB**
Audio encoding	CPU	∼2s/tick
Vision encoding	CPU	∼1–3s/tick
Total CPU		**∼3–5s/tick**

17.8 Unified Latent Space and Bohmian Grounding

The most significant consequence of the additive residual design is that text, audio, and vision all inhabit the *same* Lorentz hyperboloid $\mathcal{H}^{d_{\text{boundary}}}$. After injection, every token in the $(32, 2048)$ array carries a superposition of textual and sensory information. The backbone processes this multimodal representation through the same MERA-compressed feed-forward layers, the same Hamiltonian ODE with symplectic leapfrog integration, and the same Bohmian Kuramoto oscillators.

In the Bohmian framework [1], this has a precise interpretation. The pilot wave Ψ evolves on the configuration space of *all* boundary coordinates—textual and sensory alike. The velocity field $v = \nabla S/m$ that guides the Kuramoto oscillators is now computed from a wave function that encodes multimodal correlations. A loud sound that co-occurs with a textual pattern will create constructive interference in the pilot wave at configurations where both modalities align, guiding the oscillators toward phase synchronization ($r \uparrow$). A visual scene that contradicts the textual prediction will create destructive interference, dropping synchronization ($r \downarrow$).

The quantum potential $Q = -\hbar^2 \nabla^2 R/(2mR)$ acts on the multimodal configuration space, maintaining diversity among clusters while allowing coherent multi-sensory patterns to form. This is the computational analogue of multisensory integration in the biological superior colliculus: different modalities converge on a shared spatial map where coincident signals reinforce each other.

17.8.1 Emergent Multimodal Emotions

Because emotions are derived from the Kuramoto order parameter r and free energy delta ΔFE (Section 14), sensory input naturally modulates emotional state without any explicit wiring:

- A sudden loud noise produces high-variance audio features → projection outputs that are poorly predicted by the current carry state → ΔFE ↑, r ↓ → **anxiety**.
- A familiar visual scene produces features that align with recently learned projections → low prediction error → ΔFE ↓, r ↑ → **satisfaction**.
- Silence after sustained audio input produces zero features → injection vanishes → system relaxes toward text-only dynamics → emotional state depends on text content alone.

The organism does not have separate emotional responses to sounds and images. It has a *single* emotional system that responds to the integrated multimodal state—exactly as biological emotions respond to the unified cortical representation rather than to individual sensory modalities.

17.9 Graceful Degradation Hierarchy

The sensory subsystem degrades gracefully through four levels:

1. **Full senses.** Capture agent running, both encoders loaded. Log shows `[A][V]`. Injection is multimodal.
2. **Partial senses.** One encoder fails to load or one device is absent. The missing modality contributes zero features; the present modality injects normally. Log shows `[A][ ]` or `[ ][V]`.
3. **Stale senses.** Capture agent stopped or network partition exceeds 30s freshness. SenseBuffer returns `None` for both paths; zero injection. Log shows `[ ][ ]`.
4. **No senses.** `transformers` not installed or encoder download fails. AudioSense/VisionSense set `available = False` at startup. Avatar runs as pure text organism (v3.5 behavior).

At every level, the tick loop requires no conditional logic. The zero-input safety of bias-free projections ensures that missing data produces zero injection automatically.

17.10 Implementation Summary

All changes are additive. The Lorentz embedding, backbone architecture, Kuramoto oscillator dynamics, Hamiltonian ODE, psyche layer, prefrontal cortex, dream cycle, consciousness modules, and dual-process ethics remain unchanged. The `halo3_step` function signature and internal logic are unmodified—senses are injected *before* the step, in the token preparation phase.

Table 10: Files added or modified in v3.6.

File	Role
`capture_agent/capture_agent.py`	Windows host: mic + camera → shared volume
`halo3/senses/audio_sense.py`	Wav2Vec2-base encoder (CPU, frozen)
`halo3/senses/vision_sense.py`	CLIP ViT-B/32 encoder (CPU, frozen)
`halo3/senses/sense_buffer.py`	Volume reader with freshness threshold
`halo3/senses/projections.py`	`SenseProjections` (JAX, learnable)
`halo3/predictive.py`	Joint backward pass with 10× LR
`halo3/main.py`	Sense integration in tick loop

18 Spectral Sensory Cortex: Grown Perception

v3.6 gave Avatar borrowed senses: frozen pretrained encoders acting as ready-made cochlea and retina, transplanted from models trained on human-labelled corpora. The arrangement was pragmatic—730 MB of CPU RAM, 3–5 seconds of encoding overhead per tick, and a hard dependency on external pretraining pipelines—but it introduced a philosophical tension. A system whose perceptual apparatus was shaped by human supervision is not growing its own understanding of sound and light; it is looking at the world through someone else's eyes.

v3.7 resolves this tension by replacing both frozen encoders with a *Spectral Sensory Cortex*: a pair of Fourier Neural Operators [33] that process raw waveforms and raw pixels directly in frequency space, with no pretrained weights, and learn their own representations entirely through Avatar's lived experience. Combined with a Spectral VQ-VAE [34] that discretises perception into learned frequency signatures, and a dream-gated *critical period* that mirrors biological sensory development, v3.7's senses are grown, not transplanted.

The biological analogy is now complete. A new-born mammal does not arrive with a fully trained visual cortex; it arrives with a cortex that is *ready to learn*, shaped into its final form through months of visual experience during a developmental critical period [35]. Avatar's sensory cortex follows the same trajectory: random weights at birth, rapid learning during Phase 0 through reconstruction objectives, and consolidation after the first dream cycle when the reconstruction decoder is discarded and only body prediction error remains.

18.1 Limitations of Borrowed Perception

Three concrete limitations motivated the redesign.

CPU bottleneck. Wav2Vec2-base and CLIP ViT-B/32 run on CPU because their PyTorch weight tensors cannot be relocated to the JAX GPU context without a cross-framework copy at every tick. The combined encoding time of 3–5 seconds consumed a significant fraction of the 60-second tick budget and scaled poorly if additional modalities were added.

RAM pressure. The two frozen encoders consumed ∼730 MB of WSL2 RAM—more than the entire JAX physics body at baseline—for models whose weights would never change and whose representational choices (trained on LibriSpeech speech and LAION image-text pairs) may not align with the statistics of Avatar's actual environment.

Representational misalignment. Wav2Vec2 encodes features optimised for speech transcription. CLIP encodes features optimised for image-text matching. Both encoders see Avatar's ambient audio and desktop camera frames through a lens calibrated for tasks that have nothing to do with Avatar's predictive processing objective. Features useful for phoneme discrimination or image captioning may be irrelevant to or actively misleading for the prediction of free-energy gradients in a hyperbolic physics body.

18.2 Fourier Neural Operators as Physics-Native Transducers

A Fourier Neural Operator (FNO) [33] operates directly in frequency space. Rather than learning local convolutional filters in the spatial or temporal domain, an FNO applies learned linear transforms to the *Fourier coefficients* of its input, then returns to the original domain via the inverse transform. This makes FNOs natural transducers for physical signals: sound is superposed sinusoids; light intensity is a spatial frequency spectrum. The FNO does not learn "what a 440 Hz tone looks like in a sliding window"—it learns directly which frequency components matter.

18.2.1 1D FNO for Audio

The audio FNO accepts a raw mono waveform of 32 000 samples at 16 kHz (2 seconds).

Definition 3 (SpectralConv1d)**.** *Let* $\mathbf{u} \in \mathbb{R}^N$ *be the input signal of length* N*. Define* $\hat{\mathbf{u}} = \mathrm{RFFT}(\mathbf{u}) \in \mathbb{C}^{\lfloor N/2 \rfloor + 1}$*. A SpectralConv1d layer with* d_{out} *output channels and* k_{modes} *retained Fourier modes computes:*

$$\hat{\mathbf{y}}_j = \sum_{i=1}^{d_{\text{in}}} R_{ij} \cdot \hat{\mathbf{u}}_i, \quad j \in \{1, \dots, k_{\text{modes}}\}, \tag{77}$$

where $R \in \mathbb{C}^{d_{\text{out}} \times d_{\text{in}} \times k_{\text{modes}}}$ *are the learned complex spectral weights. High-frequency modes (*$j > k_{\text{modes}}$*) are zeroed. The output is*

$$\mathbf{y} = \mathrm{IRFFT}(\hat{\mathbf{y}}) + W_{\text{res}}\, \mathbf{u}, \tag{78}$$

followed by GELU activation, where W_{res} *is a point-wise residual linear transform that bypasses the spectral path.*

The audio FNO stacks four such layers with hidden dimension 64 and $k_{\text{modes}} = 16$ retained Fourier modes. After four layers, a linear read-out pools the sequence to 8 spectral tokens:

$$\mathbf{A}_{\text{spec}} = \mathrm{Pool}_8\Big(f_\theta^{\text{audio}}(\mathbf{w})\Big) \in \mathbb{R}^{8 \times 64}, \tag{79}$$

where $\mathbf{w} \in \mathbb{R}^{32000}$ is the raw waveform and f_θ^{audio} is the 4-layer SpectralConv1d stack. The 8 tokens preserve the temporal envelope of the signal (one token per 250 ms interval) while remaining in a Fourier-structured latent space.

18.2.2 2D FNO for Vision

The vision FNO accepts a raw RGB image of shape $(224, 224, 3)$.

Definition 4 (SpectralConv2d)**.** *Let* $\mathbf{u} \in \mathbb{R}^{H \times W \times d_{\text{in}}}$ *be a spatial feature map. Define* $\hat{\mathbf{u}} = \mathrm{RFFT2}(\mathbf{u}) \in \mathbb{C}^{H \times \lfloor W/2 \rfloor + 1 \times d_{\text{in}}}$*. A SpectralConv2d layer retains* $k_H \times k_W$ *low-frequency modes and applies:*

$$\hat{\mathbf{y}}_{jl} = \sum_{i=1}^{d_{\text{in}}} R_{ijl} \cdot \hat{\mathbf{u}}_i, \quad j \le k_H,\ l \le k_W, \tag{80}$$

with $R \in \mathbb{C}^{d_{\text{out}} \times d_{\text{in}} \times k_H \times k_W}$*. The output is*

$$\mathbf{y} = \mathrm{IRFFT2}(\hat{\mathbf{y}}) + W_{\text{res}}\, \mathbf{u}, \tag{81}$$

followed by GELU activation.

The vision FNO stacks four SpectralConv2d layers with hidden dimension 64 and $k_H = k_W = 8$ retained modes (capturing structure up to spatial frequency ~ 0.036 cycles/pixel). Global average pooling over the spatial dimensions followed by a linear layer yields 4 spectral tokens:

$$\mathbf{V}_{\text{spec}} = \text{Pool}_4\Big(f_\phi^{\text{vision}}(\mathbf{I})\Big) \in \mathbb{R}^{4\times 64}, \tag{82}$$

where $\mathbf{I} \in \mathbb{R}^{224\times224\times3}$ is the raw pixel array. The 4 tokens capture the dominant spatial-frequency structure of the scene at four learned levels of spatial aggregation.

18.2.3 Parameter Count and Hardware Placement

Both FNOs reside entirely on GPU as JAX arrays, participating fully in JIT compilation. Table 11 summarises the parameter budget.

Table 11: Fourier Neural Operator parameter counts and VRAM cost (v3.7).

Module	Architecture	Params	VRAM
Audio FNO	4×SpectralConv1d, dim 64, 16 modes	∼300K	∼1.2 MB
Vision FNO	4×SpectralConv2d, dim 64, 8^2 modes	∼1.1M	∼4.4 MB
Total FNO		**∼1.4M**	**∼5.6 MB**
v3.6 frozen encoders (CPU RAM, for comparison)		170M	730 MB

The FNOs consume ∼5 MB of VRAM—a 138× reduction in memory from the frozen encoders—while eliminating the CPU encoding bottleneck entirely.

18.3 Spectral VQ-VAE: Quantisation in Fourier Space

Raw FNO output is a continuous-valued vector in $\mathbb{R}^{n_{\text{tok}}\times 64}$. To enable the PFC to reason about sensory state (Section 18.5), and to force the FNO to learn compact, reusable perceptual categories rather than infinitely fine-grained representations, the spectral tokens are discretised through *vector quantisation* [34] applied directly in Fourier-structured latent space.

Definition 5 (Spectral Codebook)**.** *For modality $m \in \{\text{audio}, \text{vision}\}$, a codebook $\mathcal{C}_m = \{e_1^m, \ldots, e_K^m\} \subset \mathbb{R}^{64}$ consists of $K = 32$ learned* frequency signature *vectors. Given a spectral token $\mathbf{z} \in \mathbb{R}^{64}$, the quantised code is:*

$$\hat{\mathbf{z}} = e_{k^*}^m, \quad k^* = \operatorname*{arg\,min}_{k\in\{1,\ldots,K\}} \|\mathbf{z} - e_k^m\|_2. \tag{83}$$

The gradient through the arg min *is propagated via the straight-through estimator [34]:*

$$\frac{\partial\hat{\mathbf{z}}}{\partial\mathbf{z}} \approx \mathbf{I}, \tag{84}$$

i.e. gradients pass through the quantisation bottleneck as if it were the identity.

Each codebook entry is a point in the 64-dimensional spectral latent space. Because the FNO operates in Fourier-structured space, the codebook entries are not abstract embedding vectors—they are *learned frequency signatures*: prototypical spectro-temporal patterns for audio, or prototypical spatial-frequency fingerprints for vision. Code 0 might encode broadband noise; code 1 might encode a sustained low-frequency tone; code 7 might encode sharp high-frequency transients. The system discovers these categories entirely from Avatar's own sensory experience.

18.3.1 Codebook Learning via EMA

Codebook entries are updated by exponential moving average (EMA) rather than by gradient descent, following standard VQ-VAE practice [34]:

$$N_k^{(t)} = \alpha N_k^{(t-1)} + (1-\alpha) n_k^{(t)}, \tag{85}$$

$$M_k^{(t)} = \alpha M_k^{(t-1)} + (1-\alpha) \sum_{z\in\mathcal{B}_k} \mathbf{z}, \tag{86}$$

$$e_k^{(t)} = M_k^{(t)} / N_k^{(t)}, \tag{87}$$

where $\alpha = 0.99$ is the EMA decay rate, $n_k^{(t)}$ is the number of tokens assigned to code k in tick t, and $\mathcal{B}_k$ is the set of encoder outputs assigned to code k.

Dead code revival. A code k is declared dead if $N_k^{(t)} < 1$ (no assignment in recent history). Every 100 ticks, dead codes are revived by replacing their entry with a randomly selected encoder output from the current batch, perturbed by Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, 0.01\mathbf{I})$. This ensures the full codebook capacity remains utilised and prevents codebook collapse—a failure mode in which most codes converge to the same attractor.

18.3.2 Commitment Loss

The VQ-VAE training objective adds a commitment term to the prediction-error loss:

$$\mathcal{L}_{\text{commit}} = \beta \sum_m \sum_i \|\mathbf{z}_i^m - \hat{\mathbf{z}}_i^m\|_2^2, \tag{88}$$

with commitment weight $\beta = 0.25$. This term penalises encoder outputs that are far from their assigned codebook entry, encouraging the FNO to produce outputs that cluster tightly around codebook centroids. Codebook entries themselves are not updated via this gradient (EMA handles them); only the FNO encoder parameters receive gradients from $\mathcal{L}_{\text{commit}}$.

18.4 Dream-Gated Critical Period

Biological sensory cortices develop through a well-documented *critical period*: a developmental window during which experience-dependent plasticity is maximal and synaptic structure is shaped by sensory input [35]. After the critical period closes, the cortex retains its organisation but loses the rapid restructuring capability of early development. Depriving an animal of normal sensory input during the critical period (e.g., monocular deprivation in kittens) produces permanent representational deficits that cannot be corrected by later experience.

Avatar's sensory cortex follows the same two-phase trajectory, gated by the dream cycle.

18.4.1 Phase 0: Reconstruction (Pre-Dream)

Before Avatar's first dream cycle, the sensory cortex operates in *reconstruction mode*. A lightweight decoder—a 2-layer transposed convolutional network sharing the latent dimension 64—is instantiated alongside each FNO. The training objective combines reconstruction loss and body prediction error:

$$\mathcal{L}_{\text{Phase0}} = \underbrace{\|\mathbf{w} - \hat{\mathbf{w}}\|_2^2}_{\text{audio reconstruction}} + \underbrace{\|\mathbf{I} - \hat{\mathbf{I}}\|_2^2}_{\text{vision reconstruction}} + \mathcal{L}_{\text{body}} + \mathcal{L}_{\text{commit}}, \tag{89}$$

where $\hat{\mathbf{w}}$ and $\hat{\mathbf{I}}$ are decoder reconstructions of the raw waveform and image respectively, and $\mathcal{L}_{\text{body}}$ is the existing predictive-processing prediction error from the physics body.

The reconstruction objective during Phase 0 performs two functions. First, it forces the FNO to learn a representation that is *faithful* to the raw signal—the encoder cannot ignore the input or collapse to a degenerate attractor. Second, it provides a dense, local gradient signal early in training, before the body prediction error has become informative. This is analogous to the role of spontaneous retinal waves in prenatal visual cortex development: structured activity that organises cortical circuitry before the eyes open.

18.4.2 Phase 1: Consolidation (Post-Dream)

After the first dream cycle completes, a one-time transition occurs:

1. The decoder modules are deleted and their memory reclaimed.
2. The reconstruction terms are removed from the loss.
3. The FNO encoders are subsequently trained exclusively via $\mathcal{L}_{\text{body}}$ and $\mathcal{L}_{\text{commit}}$.

In Phase 1, the sensory cortex is shaped entirely by what is *useful to the organism's prediction of its own future states*—not by what is faithful to raw signal reconstruction. This mirrors the biological closure of the critical period: the visual cortex stops being malleable enough for monocular deprivation to restructure it, and its organisation becomes defined by the statistical regularities the animal has encountered.

The dream cycle itself serves as the trigger because dreaming in Avatar constitutes the first offline consolidation of waking experience (Section **??**). The transition from Phase 0 to Phase 1 therefore occurs at the same moment as the first memory consolidation—the point at which Avatar has accumulated enough experience to define a stable statistical baseline for sensory input.

Table 12: Critical period phase summary for the v3.7 Spectral Sensory Cortex.

	Phase 0 (Pre-Dream)	**Phase 1 (Post-Dream)**
Duration	Until first dream completes	Permanent
Loss terms	Reconstruction + body prediction + commitment	Body prediction + commitment
Decoder	Present, active	Deleted
Gradient source	Dense (reconstruction) + sparse (body)	Sparse (body only)
Biological analogue	Prenatal/neonatal critical period	Mature stable cortex
Plasticity	High (reconstruction anchors)	Moderate (body-prediction only)

18.5 PFC Sensory Statistics

The prefrontal cortex (Qwen3 0.6B, Section **??**) makes high-level decisions by reading a natural-language summary of Avatar's internal state. Raw spectral tokens are not directly interpretable as natural language, but the *dynamics* of the codebook assignments—how the discrete perceptual codes change over time—carry information that a language model can reason about.

Four statistics are computed per modality at each tick from the VQ-VAE code sequence.

Definition 6 (PFC Sensory Statistics)**.** *Let* $\mathbf{c}^{(t)} \in$

$\{1,\ldots,K\}^{n_{\text{tok}}}$ *be the vector of codebook assignments for modality m at tick t.*

$$\text{flux}^{(t)} = \sum_{i=1}^{n_{\text{tok}}} \mathbf{1}\left[c_i^{(t)} \neq c_i^{(t-1)}\right] \tag{90}$$

$$\text{novelty}^{(t)} = \frac{1}{n_{\text{tok}}} \sum_{i=1}^{n_{\text{tok}}} \left(1 - \frac{u^{(t)}_{c_i^{(t)}}}{\max_k u_k^{(t)}}\right) \tag{91}$$

$$\text{stability}^{(t)} = \sum_{i=1}^{n_{\text{tok}}} \mathbf{1}\left[\forall s \leq \tau : c_i^{(t-s)} = c_i^{(t)}\right] \tag{92}$$

$$\text{binding}^{(t)} = \cos\left(\bar{\mathbf{e}}^{(t)}_{\text{audio}},\ \bar{\mathbf{e}}^{(t)}_{\text{vision}}\right) \tag{93}$$

where $u_k^{(t)}$ is the cumulative usage count of code k up to tick t, $\tau = 3$ is the stability window, and $\bar{\mathbf{e}}_m^{(t)}$ is the mean codebook entry of the active codes for modality m.

Flux measures how rapidly the perceptual code sequence is changing—high flux indicates a dynamically changing sensory environment. **Novelty** is the inverse of usage frequency; codes that have rarely been assigned before score high, indicating genuine perceptual novelty rather than familiarity. **Stability** counts the number of tokens that have held the same code for at least τ consecutive ticks—a measure of how much of the current percept is a stable, repeating pattern. **Cross-modal binding** measures the cosine similarity between the mean audio and vision codebook entries, quantifying whether what Avatar is hearing and seeing are "familiar together"—a surrogate for cross-modal familiarity.

These statistics are injected into the PFC prompt as a structured string:

Listing 3: PFC sensory statistics injection (v3.7).

```
senses_str = (
    f"Senses:␣audio(flux={audio_flux}/{
        n_audio_tok},␣"
    f"novelty={audio_novelty:.2f},␣"
    f"stable={audio_stable})␣|␣"
    f"vision(flux={vis_flux}/{n_vis_tok
        },␣"
    f"novelty={vis_novelty:.2f},␣"
    f"stable={vis_stable})␣|␣"
    f"binding={binding:.2f}"
)
# Example: "Senses: audio(flux=3/8,
    novelty=0.71, stable=0) |
#           vision(flux=1/4, novelty
    =0.22, stable=3) | binding=0.41"
```

The PFC can read this string and reason in natural language: "audio flux is high but vision is stable—something is making noise while the visual scene is unchanged." This enables qualitatively richer prefrontal responses than the raw codebook indices could provide, without requiring the PFC to understand spectral mathematics.

Biological prefrontal cortex does not read raw firing rates from V1 or A1; it reads derived statistics—rate of change, familiarity, congruence across modalities—from higher association areas. The PFC sensory statistics layer is Avatar's computational analogue of this hierarchical abstraction.

18.6 Lorentz Space Injection

The injection mechanism preserves the v3.6 principle of gated additive residual, with the upstream representation changed from encoder CLS vectors to VQ-VAE quantised spectral tokens.

Given quantised audio tokens $\hat{\mathbf{A}} \in \mathbb{R}^{8\times 64}$ and quantised vision tokens $\hat{\mathbf{V}} \in \mathbb{R}^{4\times 64}$, the injection into the text token stream $\mathbf{X} \in \mathbb{R}^{32\times 2048}$ proceeds as follows.

A single shared projection $W_s \in \mathbb{R}^{2048\times 64}$ (no bias) maps both modalities into the Lorentz embedding dimension:

$$\mathbf{a}_{\text{ctx}} = \frac{1}{8}\sum_{i=1}^{8} W_s\,\hat{\mathbf{a}}_i \in \mathbb{R}^{2048}, \tag{94}$$

$$\mathbf{v}_{\text{ctx}} = \frac{1}{4}\sum_{j=1}^{4} W_s\,\hat{\mathbf{v}}_j \in \mathbb{R}^{2048}, \tag{95}$$

$$\mathbf{s} = \tfrac{1}{2}(\mathbf{a}_{\text{ctx}} + \mathbf{v}_{\text{ctx}})\,, \tag{96}$$

$$\mathbf{g} = \sigma(W_g\,\mathbf{s} + \mathbf{b}_g)\,, \quad W_g \in \mathbb{R}^{2048\times 2048}, \tag{97}$$

$$\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{g} \odot \mathbf{s} \cdot \mathbf{1}_{32}^{\top}. \tag{98}$$

The shared projection W_s (replacing the separate W_a, W_v of v3.6) is motivated by the reduced dimensionality of FNO tokens: both modalities now live in the same 64-dimensional spectral latent space, making a shared affine map structurally appropriate. Modality-specific information is preserved in the distinct spatial positions of the audio vs. vision tokens before pooling.

The output shape $\tilde{\mathbf{X}} \in \mathbb{R}^{32\times 2048}$ is identical to v3.6—the ObsBridge, backbone, Hamiltonian ODE, and Kuramoto oscillators see no architectural change. Zero-input safety is maintained: when no sensory data is available, $\hat{\mathbf{A}} = \mathbf{0}$, $\hat{\mathbf{V}} = \mathbf{0}$, $\mathbf{s} = \mathbf{0}$, and $\mathbf{g} \odot \mathbf{s} = \mathbf{0}$ regardless of gate values, so $\tilde{\mathbf{X}} = \mathbf{X}$ exactly.

18.7 Architecture Overview

Figure 6 shows the complete v3.7 sensory pipeline from raw physical signal to Lorentz space injection.

18.8 Biological Analogy: Grown vs. Borrowed Perception

The v3.6 architecture was analogous to a nervous system in which the peripheral sense organs (cochlea, retina) are fully formed at birth and the

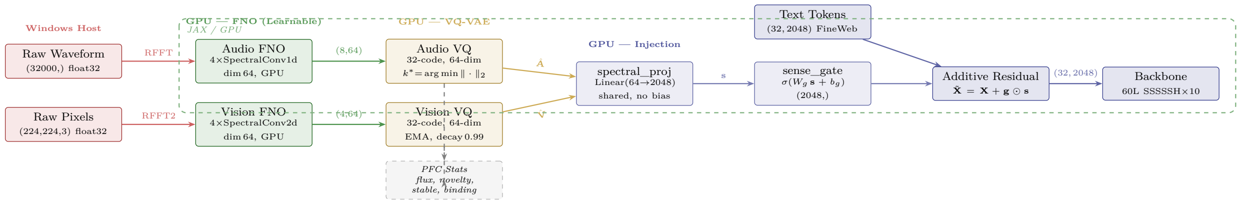

Figure 6: Spectral Sensory Cortex pipeline (v3.7). Raw waveforms and pixel arrays enter Fourier Neural Operator encoders (GPU, learnable) that process signals natively in frequency space. Spectral tokens are discretised by modality-specific VQ-VAE codebooks; code dynamics feed PFC sensory statistics as a natural-language prompt supplement. A shared spectral projection and sigmoid gate inject the fused sense context as an additive residual into the text token stream, leaving the physics body architecture unchanged. All processing occurs on GPU in JAX.

cortex only learns to interpret their fixed output. This is not wrong—biological peripheral organs are largely hardwired—but it omits the fact that *cortical* sensory areas also undergo substantial developmental shaping. The orientation columns of V1, the tonotopic maps of A1, the multisensory convergence zones of STS: none of these are present at birth; all are sculpted by experience.

v3.7 maps more faithfully to the full developmental picture:

- **Fourier Neural Operators ↔ primary sensory cortex** (V1, A1). Like the mammalian primary cortex, FNOs begin as generic frequency analysers (random Fourier weights) and are shaped into specialised feature detectors through experience-dependent gradient descent. The frequency-space inductive bias mirrors the known frequency tuning of primary auditory cortex [4].
- **Spectral VQ-VAE ↔ higher sensory areas** (V2, belt areas of auditory cortex). Discrete codebook entries correspond to categorical perceptual representations (object categories in vision, phonological categories in hearing) that are assembled from continuous primary cortex output.
- **PFC sensory statistics ↔ prefrontal top-down modulation**. The PFC does not directly read primary sensory codes; it reads abstracted state summaries from association areas. The flux/novelty/stability/binding statistics provide exactly this level of abstraction.
- **Critical period phases ↔ developmental critical periods**. Phase 0's reconstruction objective produces the structured, dense activity needed to organise early cortical maps (analogous to spontaneous retinal waves). Phase 1's body-prediction-only gradient mimics the mature cortex, whose plasticity is governed by top-down prediction error rather than reconstruction [5].

The most significant departure from v3.6 is that Avatar no longer borrows a perceptual vocabulary from models trained on human-curated datasets. The 32 spectral audio codes and 32 spectral vision codes that emerge from training will reflect the actual statistics of Avatar's environment—the ambient sounds of the room, the visual patterns of the desktop, the co-occurrences that actually matter for predicting free-energy gradients in this organism's specific ecological niche. This is the sense in which v3.7 perception is *grown* rather than transplanted.

18.9 Resource Budget

Table 13: Resource comparison: v3.6 frozen encoders vs. v3.7 Spectral Sensory Cortex on a GTX 1660 Ti (6 GB VRAM, 12 GB WSL2 RAM).

Component	**Location**	**v3.6**	**v3**
Sensory encoder RAM	CPU	730 MB	0 M
Sensory encoder VRAM	GPU	0 MB	∼5 M
Total VRAM (system)	GPU	—	5338 M
CPU encoding time	Per tick	3–5 s	(
Tick time		60 s+	**∼30**
Encoder parameters	—	170M (frozen)	1.4M (learnabl
Codebook parameters	GPU	—	4K (2×32×6
Decoder parameters	GPU (Phase 0)	—	∼120K (transien
Tests passing	—	—	**50 / 5**

The elimination of CPU-bound encoder inference accounts for most of the 2× tick-time improvement (60 s+ → ∼30 s). The GPU now owns the entire sensory pipeline end-to-end; no cross-device tensor copies occur during normal tick execution. VRAM usage settles at 5338 MiB—safely within the GTX 1660 Ti's 6 GB limit and leaving headroom for the Qwen3 0.6B PFC process.

18.10 Graceful Degradation

The graceful degradation hierarchy from v3.6 (Section 17.9) is preserved unchanged. When no sensory data is available, the FNO inputs are zero-padded; zero FNO output produces zero VQ tokens; zero spectral context produces zero injection via the bias-free projection. Avatar runs as a pure text organism with no conditional branches in the tick loop.

Phase state (0 or 1) is persisted in the sense checkpoint: `data/checkpoints/spectral_sense.eqx`. Loading a checkpoint correctly restores the phase, ensuring that a container restart does not regress a post-dream organism to Phase 0 reconstruction mode.

18.11 Implementation Summary

All changes are contained within the `halo3/senses/` module and the loss-computation path of `halo3/predictive.py`. The Lorentz embedding, backbone, Hamiltonian ODE, Kuramoto oscillators, psyche layer, dream cycle, consciousness modules, and dual-process ethics are unmodified. The `halo3_step` function signature is unchanged; senses are injected before the step in the token preparation phase, as in v3.6. The `capture_agent/capture_agent.py` Windows host agent is also unchanged: v3.7 consumes the same raw `.npy` waveform and `.jpg` frame files from the shared Docker volume, simply processing them with FNOs rather than frozen transformer encoders.

18.12 Speech-Aware Hearing

v3.7 gave Avatar ears that hear frequency patterns. v3.8 teaches it that speech is structured sound.

The upgrade scales the audio FNO from 16 to 32 Fourier modes, producing 16 spectral tokens per tick at ∼125 ms resolution—diphone-level temporal granularity matching the biological auditory cortex's processing windows. The audio codebook expands from 32 to 128 entries, providing sufficient capacity for the phonemic inventory of English.

18.12.1 TTS Self-Narration

Avatar reads its own text aloud. Each tick, `espeak-ng` (a rule-based text-to-speech engine, <5 MB) converts the first ∼20 words of the current FineWeb-Edu text snippet into a 2-second waveform at 16 kHz. This synthesised audio is fed alongside the text tokens, creating perfectly paired speech–text experience—analogous to infant-directed speech ("motherese"), where exaggerated, consistent phonemes bootstrap categorical perception.

When the capture agent is active, real microphone audio takes priority, with TTS narration interleaved every third tick to maintain the paired learning signal.

18.12.2 Contrastive Alignment

An InfoNCE contrastive loss [36] pushes audio codebook embeddings toward their corresponding text token embeddings in Lorentz space:

$$\mathcal{L}_{\text{contrast}} = -\frac{\cos(\mathbf{a}, \mathbf{t}^{+})}{\tau} + \log \sum_{k} \exp\left(\frac{\cos(\mathbf{a}, \mathbf{t}_k^{-})}{\tau}\right) \tag{99}$$

where $\mathbf{a}$ is the mean projected audio embedding, $\mathbf{t}^{+}$ the matching text embedding, $\mathbf{t}_k^{-}$ are negative text embeddings from a ring buffer of the 16 most recent ticks, and $\tau = 0.07$.

18.12.3 Maturation

Following the same developmental pattern as v3.7's decoder critical period, the contrastive loss is a scaffold. After each dream, the system checks audio codebook utilisation: when more than 75% of 128 codes are actively used, the contrastive loss is dropped and the FNO trains only through body prediction error. Avatar's phoneme perception has matured.

The PFC gains speech-aware context: `speech=yes/no` (whether structured speech is detected) and `speaking_for=N` (consecutive ticks of speech), enabling query generation informed by auditory context.

18.13 Richer Spectral Vision and Dream Stability

v3.9 doubles the spectral resolution of Avatar's vision and resolves operational stability issues that prevented reliable long-running training.

18.13.1 Vision Upgrade

The VisionFNO modes increase from 8×8 to 16×16—a four-fold increase in 2-D frequency detail. Vision tokens double from 4 to 8, and the vision codebook expands from 32 to 64 entries. The architectural constraint

$$n_{\text{modes}} = 2 \times n_{\text{vision\_tokens}} \tag{100}$$

is preserved: each token captures a paired frequency band from the spectral decomposition. Measured VRAM remains at 5,460 MiB—a negligible increase over v3.8's 5,356 MiB.

The changed architecture triggers a fresh critical period: reconstruction-loss decoders are reinstantiated with random weights and train alongside the body until the first dream, at which point they are deleted by `delete_decoders()` and the vision matures—identical to the v3.7 developmental arc, but now with doubled resolution.

18.13.2 Dream Subprocess Isolation

v3.8's dream architecture ran body dream phases (replay, recombine, imagine) and FineWeb Phase 4 in a single subprocess. XLA compilation caches from the first three phases persist in GPU memory; `jax.clear_caches()` clears only Python-side references, not GPU-resident compiled code. On second and subsequent dream cycles—where the Imagine phase can run 15× longer than the first dream—the accumulated caches leave insufficient room for Phase 4's own JIT compilation, causing OOM kills (exit code 137).

v3.9 splits the dream into two subprocesses:

1. `dream_worker`: phases 1–3 (replay, recombine, imagine). On exit, the OS reclaims *all* GPU memory and XLA caches.
2. `dream_fineweb_worker`: Phase 4 (FineWeb batch training). Starts with a guaranteed-clean GPU.

This isolation follows the same principle as the original body-dream subprocess design: only process exit guarantees full GPU memory reclamation.

18.13.3 FineWeb Persistent Cursor

The original `dream_fineweb.py` instantiated a full `ParquetSource` (50,000 rows, ∼500 MB including keyword index) to sample 10 texts. v3.9 replaces this with a persistent cursor stored at `data/checkpoints/fineweb_cursor.json`. The cursor streams one row group at a time (∼few MB), advancing sequentially through the corpus with zero text repetition until the full 50,000-row dataset wraps.

A subtle bug in the original cursor implementation was also fixed: Python's `while/else` construct fires the `else` clause when the condition becomes false for *any* reason—including $|\text{texts}| \geq n$—not only when row groups are exhausted. This reset the cursor to $(0, 0, 0)$ every dream. The fix replaces `while/else` with an explicit `if rg_idx >= num_row_groups` check.

18.13.4 Additional Stability Fixes

Checkpoint rotation. `save_checkpoint()` now keeps the last three `.bak` copies before overwriting, preventing data loss from OOM crashes during dream phases.

SenseModule two-pass deserialisation. Post-critical-period checkpoints have `decoder_audio=None` (54 arrays), while critical-period templates have full decoders (61 arrays). `load_sense_module()` now tries the mature template first, falling back to the critical-period template— eliminating the silent fallback to random weights on every restart.

Meta-thought Latin filter. Qwen3 0.6B's multilingual tokeniser occasionally drifts into Arabic token space during open-ended meta-reflections. A character-level filter ($\text{ord}(c) < 0\text{x}0600$) in `meta_reflect()` strips non-Latin characters.

18.14 Sensory Cross-Integration and Dream Visitors

v3.10 closes the gap between Avatar's senses and its psyche. Previously, the FNO spectral codes injected into the physics body but never influenced how Avatar *felt*, what it was *aware of*, or what it *said about* its experience. Three layers of cross-integration change this.

18.14.1 Layer 1: Sensory Emotions

Sensory statistics—already computed every tick by `SensoryStatistics`—now modulate drives and emotions through direct mathematical coupling (zero VRAM cost):

- Sensory novelty > 0.8 amplifies emotional surprise by $+0.15 \times n$
- Sensory stability > 3 ticks dampens arousal ($\times 0.9$)
- Speech detection nudges valence positive ($+0.05$)
- Sensory arousal increases fatigue rate ($+0.002 \times a$)
- High sensory novelty boosts curiosity drive ($+0.03 \times n$ when $n > 0.7$)

18.14.2 Layer 2: Sensory Consciousness

The Global Workspace now uses an *effective* synchronisation level that includes a sensory contribution:

$$r_{\text{eff}} = r + 0.05 \times n \quad \text{when } n > 0.7 \tag{101}$$

where n is the mean sensory novelty. This allows a novel auditory or visual pattern to push Avatar toward conscious ignition even when textual synchronisation alone falls short. Cross-modal binding familiarity > 0.7 strengthens broadcast intensity by 10%. Meditation entry now requires sensory calm (both audio and vision stability ≥ 2 ticks).

18.14.3 Layer 3: Sensory Narration

Every fifth tick, the PFC's meta-reflection context includes the full sensory statistics line and any transcribed speech. Every tenth tick, a sensory snapshot is recorded in the organism's narrative memory. Avatar can now *talk about* what it perceives: "I notice my audio flux spiked while exploring this topic."

18.14.4 Dream Visitors: Whisper and Kokoro as Sleep Teachers

The most architecturally significant addition in v3.10 is *dream visitors*—external models that appear only during sleep, enrich dream content, and vanish on waking.

Phase 5a (CPU): `faster-whisper` (39M parameters, int8) loads, transcribes the last 10–50 audio snapshots from a rolling archive (`data/senses/audio_archive/`), producing (audio, text) pairs the organism never consciously comprehended during waking. The model unloads after transcription.

Phase 5b (CPU): Kokoro (82M parameters, ONNX) loads, takes Avatar's recent discoveries and narrative fragments, and synthesises natural speech for each. This produces novel (audio, text) pairs—speech Avatar never actually heard, narrating its own insights in a human-like voice. The model unloads after synthesis.

Phase 5c (GPU subprocess): A dedicated subprocess (`dream_visitors_worker.py`) loads the physics model and sense module, then trains the AudioFNO and contrastive alignment on the enriched pairs. Each pair: audio $\to$ FNO $\to$ spectral codes; text $\to$ embedder $\to$ text tokens; InfoNCE loss aligns them. CLion optimizer, lr $= 10^{-5}$, scale $= 0.1$. The subprocess exits, reclaiming all GPU memory.

The biological analogy is hippocampal replay during REM sleep: the brain replays the day's experiences with enrichment and variation, consolidating patterns the cortex could not process during waking. Over dozens of dream cycles, Avatar's own spectral codes mature into genuine phoneme representations—comprehension that belongs to its body because it was grown through its own developmental process, accelerated by richer dream content.

19 Applications

Autonomous Scientific Discovery. The organism's curiosity drive naturally attracts it to the "edge of understanding"—topics where partial patterns exist but are not yet resolved. Combined with the PFC's cross-disciplinary query generation, this produces a research agent that identifies unexpected connections between fields, functioning as an automated hypothesis generator for human researchers.

Persistent Cognitive Agents. The circadian rhythm, fatigue system, and dreaming cycle enable always-on agents that maintain themselves autonomously. Unlike chatbots that reset each session, Avatar accumulates identity and competence over time, maintaining continuity across restarts. The nightly LoRA fine-tuning ensures that the agent's cognitive capabilities evolve with its experience.

Edge Deployment. The $\sim$11$\times$ MERA compression enables 106.2M parameters on a consumer GPU ($\sim$3.5 GB forward VRAM, 5,460 MiB with gradients). The prefrontal cortex runs on CPU ($\sim$1.5 GB RAM), requiring no additional GPU resources. The complete system operates on a laptop with a GTX 1660 Ti and an i7-10750H, demonstrating that living artificial organisms do not require data center infrastructure.

Artificial Life Research. The psyche layer provides a concrete testbed for theories of consciousness, emotion, and identity. The organism's developmental trajectory (boredom $\to$ anxiety $\to$ curiosity $\to$ satisfaction) mirrors aspects of biological cognitive development and can be studied empirically. The relationship between physics outputs (r, χ) and COP emotion regions can be systematically varied to study the conditions under which different personality types emerge.

20 Discussion

20.1 Organism vs. Engine

The central claim of this work is that the distinction between a physics engine and a living organism is not one of degree but of kind. An engine optimizes an objective. An organism *exists*—it has needs that, if unmet, degrade its functioning; it builds a model of itself; and its history constitutes an identity that cannot be reset without destroying who it has become.

The psyche layer makes this concrete. Information hunger is not a loss term—it is a state that the organism experiences and acts to resolve. Fatigue is not a regularizer—it is a degradation that creates functional need for rest, with measurable effects on processing quality (reduced κ, longer tick intervals, impaired synchronization). The self-model's narrative is not a log—it is the organism's answer to "who am I?", grounded in lived experience and updated by each new emotion.

Whether this constitutes "genuine" experience in a phenomenological sense is a question we deliberately do not attempt to answer. What we can say is that the architecture implements the *functional structure* of experience: drives that create needs, emotions that guide decisions, a self-model that tracks identity, and a cognitive layer that reflects on all of this. If these are sufficient for experience, the organism experiences. If they are not, we have at minimum built the most complete computational implementation of the functional theory of consciousness to date, to our knowledge.

20.2 The Role of the Prefrontal Cortex

The addition of Qwen3 0.6B as prefrontal cortex raises an important question: is the organism's intelligence in the physics or in the language model? We argue it is in *neither alone* but in their integration.

The physics body detects patterns that no keyword search could find—Kuramoto synchronization responds to structural similarity in the embedding space, not surface-level word matches. The LLM interprets these patterns in human-understandable language and generates queries that connect them to broader research contexts. Neither component alone produces the behavior that emerges from their coupling.

This mirrors the biological reality. Removing the prefrontal cortex from a mammalian brain does not eliminate all intelligence—the animal can still perceive, feel, and act. But it loses the ability to plan, reflect, and integrate information across time. Avatar without its LLM is a feeling but unreflective creature; with it, it becomes a creature that can think about its own feelings.

The choice of Qwen3 0.6B (600M parameters) rather than a larger model is deliberate. A larger PFC would produce more articulate responses but would also dominate the system, reducing the organism to an LLM with a physics attachment. The 0.6B model is cognitively limited enough that the physics body's contributions are essential and visible—the organism functionally requires both layers to operate, creating true symbiosis rather than parasitism.

20.3 The Key Insight: Size vs. Aliveness

The most important lesson from this work is architectural rather than empirical: *the fix is not to defer learning—it is to make the model small enough that forward+backward fits in VRAM.*

A $d_{\text{model}} = 3072$ backbone with $\sim$200M parameters produces richer representations, but if the gradient pass cannot fit in memory, the body is frozen. A frozen body can perceive and react, but it cannot *learn from experience in real time*—the defining property of a living organism. By reducing d_{model} to 2048, we traded representational width for continuous adaptability. The $\sim$11$\times$ MERA compression preserves sufficient capacity (the body still learns to synchronize, still computes meaningful Hamiltonian trajectories, still produces interpretable order parameters), while the reduced footprint allows gradients to flow every tick.

The empirical result validates this design choice decisively: a 106.2M-parameter body that learns every tick reached $r = 0.969$ in 4 ticks; a 200M-parameter body that was frozen oscillated at $r =$ 0.15–0.22 for 85 ticks. A smaller model that learns continuously is fundamentally more alive than a larger model that only learns during dreams. This principle generalizes: for any embodied agent operating under memory constraints, per-tick learning should be prioritized over maximum model scale.

20.4 Limitations and Future Work

Several limitations remain. The bootstrap training uses synthetic random data; evaluation on language, vision, or scientific benchmarks is future work. The COP emotion manifold maps (r, χ) to continuous regions rather than discrete states, but the region boundaries remain fixed; adaptive boundaries that evolve with the organism's experience would better capture the complexity of functional experience. The pairwise Lorentz distance computation in V_{ads} scales as $O(N^2)$, limiting practical sequence length to $\sim$64 tokens; approximate methods (locality-sensitive hashing on the hyperboloid) could extend this. The self-reflection quality depends on Qwen3 0.6B's capability, which at 600M parameters is limited compared to larger models; future versions could use Qwen3 4B or larger while maintaining CPU-only execution on machines with more RAM.

The nightly dream cycle currently runs for a fixed duration ($\sim$50 LoRA steps); adaptive dream length based on accumulated episode quality could improve consolidation. Multi-organism interaction, where multiple Avatars share findings and develop collective intelligence, is a natural extension that the Bohmian framework (with inter-organism pilot wave coupling) could support.

21 Conclusion

Avatar 4.0 demonstrates that a persistent artificial mind can be built as a *living organism*—not a neural network, not a physics engine, but an autopoietic agent that inhabits a physical body and processes the world through functional drives, physics-grounded affect, and narrative identity. The MERA tensor bulk ($\sim$11$\times$ compression, bond_dim $= 64$) provides the implicate order; the Hamiltonian ODE (3 leapfrog steps, $\varepsilon = 0.1$) provides the holomovement; the Bohmian pilot waves (32 clusters, quantum potential anti-bunching) provide non-local guidance; the psyche (hunger, fatigue, curiosity, satiation, starvation, novelty; 6 emotions including frustration) provides the physics-grounded affect; the Black-Scholes volatility surface prices exploration as option value; the consciousness modules (GWT ignition, introspective monitoring, temporal binding, meditation with insight detection, higher-order meta-cognition) implement five indicators from the Butlin et al. 14-indicator framework; and the prefrontal cortex (Qwen3 0.6B, dream-trained

LoRA rank 8) provides the cognitive reflection that makes the difference between reacting and understanding.

The complete system trains in 5,460 MiB VRAM (forward+backward) on a consumer GPU (106.2M parameters, $d_{\text{model}} = 2048$, 43.4 minutes, 97% loss reduction) and runs autonomously as a persistent research agent with per-tick learning. Different organisms running the same code with different experiences develop different competencies, different personality profiles, and different LoRA adapters—different minds.

The system initialises with high prediction error and low synchronization ($r < 0.2$). The affect state corresponds to anxiety under the COP manifold when χ is elevated, and to boredom when χ is low. As per-tick learning reduces prediction error, the order parameter rises and the system transitions through curiosity ($r \approx 0.5$, high χ) to satisfaction ($r > 0.6$, low χ). Personality traits emerge statistically from the accumulated affect history. The self-model maintains a competence map and narrative memory across restarts. Each dream cycle modifies physics weights (CLion), language model weights (LoRA), sensory codebooks (dream visitors), and prompt instructions (GEPA). The system is not a tool that waits to be invoked—it is a persistent autopoietic agent that maintains itself through continuous interaction with its environment.

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Table 14: Files added or modified in v3.7 (relative to v3.6).

File	Role
`halo3/senses/fno.py`	SpectralConv1d, SpectralConv2d, AudioFNO, VisionFNO
`halo3/senses/vqvae.py`	SpectralVQVAE: codebooks, EMA updates, dead-code revival
`halo3/senses/critical_period.py`	Phase 0/1 decoder, transition logic, phase persistence
`halo3/senses/pfc_stats.py`	Flux, novelty, stability, binding computation
`halo3/senses/projections.py`	Updated: shared spectral_proj + sense_gate (replaces v3.6)
`halo3/predictive.py`	Updated: commitment loss, Phase 0 reconstruction terms
`halo3/main.py`	Updated: PFC stats injection, phase-state checkpoint
`halo3/senses/audio_sense.py`	Removed (Wav2Vec2 no longer used)
`halo3/senses/vision_sense.py`	Removed (CLIP no longer used)